

# ACTSC 372 Investment Science and Corporate Finance

University of Waterloo - Fall 2025

10th November 2025

Instructor: Surya Banerjee

L<sup>A</sup>T<sub>E</sub>X: Xing Liu

## Contents

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| <b>0</b> | <b>The Role of Financial Intermediaries</b>   | <b>3</b>  |
| <b>1</b> | <b>Investment Rules</b>                       | <b>4</b>  |
| 1.1      | The NPV Rule . . . . .                        | 4         |
| 1.2      | Review of ACTSC 231 . . . . .                 | 6         |
| 1.3      | Other Investment Rules . . . . .              | 7         |
| 1.4      | Equivalent Annual Costs (EAC) . . . . .       | 14        |
| <b>2</b> | <b>Capital Budgeting</b>                      | <b>15</b> |
| 2.1      | Rules for Calculating NPVs . . . . .          | 15        |
| 2.2      | Capital Budgeting Under Uncertainty . . . . . | 20        |
| 2.3      | Utility Theory . . . . .                      | 22        |
| 2.4      | Risk Aversion . . . . .                       | 24        |
| 2.5      | Mean-Variance Utility . . . . .               | 28        |
| <b>3</b> | <b>Capital Asset Pricing Model (CAPM)</b>     | <b>30</b> |
| 3.1      | Risk and Return . . . . .                     | 30        |
| 3.2      | Capital Asset Pricing Model . . . . .         | 33        |
| 3.3      | Security Market Line (SML) . . . . .          | 49        |

|          |                                                   |           |
|----------|---------------------------------------------------|-----------|
| <b>4</b> | <b>Arbitrage Pricing Theory (APT)</b>             | <b>57</b> |
| 4.1      | APT and Factor Models . . . . .                   | 57        |
| 4.2      | Factors Affecting Beta . . . . .                  | 62        |
| 4.3      | Weighted Average Cost of Capital (WACC) . . . . . | 66        |
| <b>5</b> | <b>Efficient Market Hypothesis (EMH)</b>          | <b>68</b> |
| <b>6</b> | <b>Capital Structure</b>                          | <b>74</b> |
| 6.1      | The Basics . . . . .                              | 74        |
| 6.2      | Modigliani Miller Propositions . . . . .          | 77        |
| 6.3      | Limits to Debt . . . . .                          | 83        |
| <b>A</b> | <b>Tutorials</b>                                  | <b>85</b> |
| A.1      | Tutorial 1 (2025/09/19) . . . . .                 | 85        |
| A.2      | Tutorial 2 (2025/10/03) . . . . .                 | 87        |

## 0 The Role of Financial Intermediaries

---

Lecture 1, 2025/09/03

---

What is financial investment?

- Ownership / loan contracts helped by financial intermediaries.

### Role of Financial Intermediaries

- **Size intermediation:** the bank gets small lenders together with the “big” borrowers.
- **Term intermediation:** between short term lenders and long term borrowers.
- **Risk intermediation:** pooling resources reduces individual risk.

**Consumer Smoothing:** see handwritten notes.

# 1 Investment Rules

## 1.1 The NPV Rule

**Question:** Given a choice of projects, what project should a rational investor choose?

**Definition 1.1 (Project).**

A **project** is a collection of cash flows  $\{C_0, C_1, C_2, \dots\}$  where  $C_t$  = cash flow at time  $t$ .

*Note.*

- $C_t < 0 \implies$  cost.
- $C_t > 0 \implies$  revenue.

**Definition 1.2 (Pure Investment Project).**

A **pure investment project** is a project where  $C_0 < 0$  and  $C_t \geq 0$  for all  $t > 0$ .

**Definition 1.3 (Pure Financing Project).**

A **pure financing project** is a project where  $C_0 > 0$  and  $C_t \leq 0$  for all  $t > 0$ .

**Definition 1.4 (Independent Projects).**

$A$  and  $B$  are **independent projects** if the cash flows of  $A$  do not affect the cash flows of  $B$ .

*Note.* Between projects  $A$  and  $B$ , we can choose:

- one,
- both,
- or neither.

**Definition 1.5 (Mutually Exclusive Projects).**

$A$  and  $B$  are **mutually exclusive projects** if choosing  $A \implies$  we cannot choose  $B$  and vice versa (we cannot choose both).

**Example.** Given  $C_0, C_1, C_2, \dots$ . How to decide if this is a “good” project?

*Proof.* See NPV rule below. □

**Definition 1.6 (Net Present Value (NPV)).**

The **NPV** of a project is

$$NPV = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \dots = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

where units = dollars.

**Definition 1.7 (NPV Rule).**

- For independent projects, choose all projects with  $NPV > 0$ .
- For mutually exclusive projects, choose the project with the highest  $NPV$ , as long as  $NPV > 0$ .

**Example.** Let  $C_0 = -100, C_1 = 110, C_t = 0$  for  $t \geq 2$ , and  $r = 8\%$ . Then,

$$NPV = -100 + \frac{110}{1.08} > 0.$$

So, we should accept this project.

**Example.** Let  $C_0 = 100, C_1 = -110$ , and  $r = 8\%$ . Then,

$$NPV = 100 - \frac{110}{1.08} < 0.$$

So, we should reject this “bad” project.

What is  $r$ ?  $\implies r$  = “opportunity cost” of the dollar.

**Advantages of NPV rule:**

- NPV takes into account the time value of money.
- NPV takes into account cash flows, not accounting incomes.

**Example.** Let  $C_0 = 100$ ,  $C_1 = 115$ ,  $r = 10\%$ . The project has positive NPV and the owner will be happy if the project is taken.

**Disadvantages of NPV rule:**

- The cash flows are uncertain.
- What should be the right “ $r$ ”? (see later: CAPM)

## 1.2 Review of ACTSC 231

Some important cash flows to recall from ACTSC 231:

- Perpetuity.
- Perpetuity with growth.
- Annuity.

**Definition 1.8 (Perpetuity, Perpetuity Factor).**

A **perpetuity** is an annuity with no end dates (infinite # of payments). Let  $C_t = C$  for all  $t \geq 1$ . Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots \\ &= \frac{C}{1+r} \left[ 1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots \right] \\ &= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1}{1+r}} \\ &= \frac{C}{r}. \end{aligned}$$

The **perpetuity factor** is  $\frac{1}{r}$ .

**Definition 1.9 (Perpetuity with Growth).**

Let  $C_t = C(1 + g)^{t-1}$  for  $t \geq 1$ . Then,

$$\begin{aligned}
 PV &= \frac{C}{1+r} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \dots \\
 &= \frac{C}{1+r} \left[ 1 + \frac{1+g}{1+r} + \left( \frac{1+g}{1+r} \right)^2 + \dots \right] \\
 &= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1+g}{1+r}} \\
 &= \frac{C}{r-g}, \quad g < r.
 \end{aligned}$$

**Definition 1.10 (Annuity, Annuity Factor).**

Let  $C_t = C$  for  $1 \leq t \leq n$ . Then,

$$\begin{aligned}
 PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \dots + \frac{C}{(1+r)^n} \\
 &= \frac{C}{1+r} \left[ 1 + \frac{1}{1+r} + \dots + \frac{1}{(1+r)^{n-1}} \right] \\
 &= \frac{C}{1+r} \cdot \frac{1 - \left( \frac{1}{1+r} \right)^n}{1 - \frac{1}{1+r}} \\
 &= C \cdot \frac{1 - (1+r)^{-n}}{r}.
 \end{aligned}$$

The **annuity factor** is  $\frac{1}{r} \left[ 1 - \left( \frac{1}{1+r} \right)^n \right]$ .

**1.3 Other Investment Rules****Definition 1.11 (Payback Period).**

The **payback period** is the # of years required to recoup the initial investment (ignoring the time value of money).

**Example.** Let  $C_0 = -300$ ,  $C_t = 100$  for  $t \geq 1$ . The payback period is 3 years.

*Note.* The units of payback period is years.

**Definition 1.12 (Payback Period Rule).**

Decide on a cutoff period,  $T^*$ . If the payback period  $\leq T^*$ , accept; otherwise, reject it.

**Disadvantages of Payback Period Rule:**

- It does not take into account the time value of money.
- The cutoff is arbitrary.
- Negative NPV may be accepted.
- Positive NPV may be rejected.

**Definition 1.13 (Discounted Payback Period).**

The **discounted payback period** is the payback period, but without ignoring the time value of money.

**Example.** Let  $C_0 = -300$ ,  $C_t = 100$  for  $t \geq 1$ , and  $r = 12.5\%$ . We are solving  $\sum_{t=1}^T \frac{C_t}{(1+r)^t} = -C_0$ . This implies that the discounted payback period,  $T$ , is 4 years.

**Definition 1.14 (Internal Rate of Return (IRR)).**

The **internal rate of return (IRR)** of a project is the cost of capital such that the NPV is zero.

**Example.** Let  $C_0 = -100$ ,  $C_1 = 112$ ,  $r = 10\%$ . We have two ways to determine if we should accept this project:

- (1) Calculate NPV:  $NPV = -100 + \frac{112}{1.1} > 0 \implies$  accept.
- (2) The project has a return of 12%. Since it is bigger than  $r = 10\%$ , we should accept it.

**Question:** How to solve for the IRR?

**Answer:** Calculate the NPV and equal it to zero, then solve for  $r^*$ . That is,

$$\sum_{t=0}^T \frac{C_t}{(1+r^*)^t} = 0.$$



**Definition 1.15 (IRR Rule).**

Calculate the IRR of a project.

- For independent projects, choose all projects with  $IRR > r$ .
- For mutually exclusive projects, choose the project with the highest IRR (as long as  $IRR > r$ ).

**Advantages of IRR rule:**

- In order to calculate IRR, we do not need  $r$ .
- It usually agrees with the NPV rule.

**Disadvantages of IRR rule:**

The first two disadvantages are for independent projects.

- (1) IRR cannot distinguish between investment and financing.

*Note.*  $(-1) \sum \frac{C_t}{(1+r^*)^t} = 0(-1)$ , the solution remains the same.

**Example.** Let  $C_0 = -100, C_1 = 112, r = 10\%$ . The IRR is 12% and we should accept this project.

Let  $C_0 = 100, C_1 = -112, r = 10\%$ . The IRR is still 12% but we should reject this project.

- (2) Polynomial equation problem: finding the IRR = solving a polynomial equation, i.e.

$$C_0 + \frac{C_1}{1+r^*} + \frac{C_2}{(1+r^*)^2} + \cdots + \frac{C_T}{(1+r^*)^T} = 0.$$

However, we could have no roots, multiple roots, imaginary roots, or complex roots.

**Example.** Let  $C_0 = 0, C_1 = 100, C_t = 0$  for all  $t \geq 2$ . The  $IRR = \infty$ .

**Example.** Let  $C_0 = -100, C_1 = 230, C_2 = -132$ . Then,

$$-100 + \frac{230}{1+r^*} - \frac{132}{(1+r^*)^2} = 0 \implies r^* = 10\% \text{ or } 20\%.$$

This is problematic, since which IRR should we use?

We will need the following result to deal with the multiple IRR problem.

**Theorem 1.1 (Descartes' Rule of Signs).**

The # of positive roots of a polynomial is  $\leq$  # of “sign switches” of the coefficients.

**Example.** For the previous example where  $C_0 = -100, C_1 = 230, C_2 = -132$ , there are 2 sign switches. Thus, there are at most 2 positive roots.

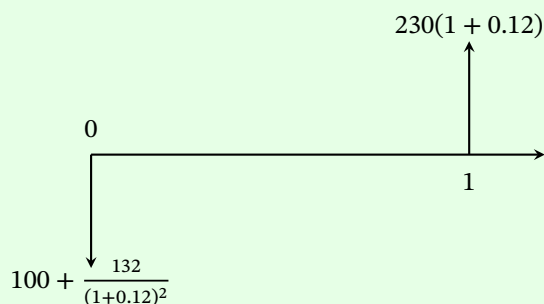
We have the following MIRR to avoid multiple solutions.

**Definition 1.16 (Modified Internal Rate of Return (MIRR)).**

The **modified IRR**

- brings all negative cash flows to time  $t = 0$ , and
- brings all positive cash flows to time  $t = T$ ,

**Example.** Let  $C_0 = -100, C_1 = 230, C_2 = -132$ , and  $r = 12\%$ . Then,



Calculate the IRR of the modified project (MIRR):

$$-\left(100 + \frac{132}{(1 + 0.12)^2}\right) + \frac{230(1 + 0.12)}{1 + r^*} = 0 \implies r^* = 12.36\%.$$

*Note.* We only have 1 “sign switch” in the modified project, so MIRR is unique. Then, we can compare MIRR with  $r$ , and choose the project with  $\text{MIRR} > r$ . But we need to know today’s interest rate  $r$ .

**Disadvantages of IRR rule (continued):**

For mutually exclusive projects (i.e. we can only choose one project.):

(3) Scaling problem: the IRR is biased towards “smaller” projects.

**Example.**

Project A:  $C_0 = -1, C_1 = 2$ .

Project B:  $C_0 = -100, C_1 = 150$ .

Note that  $IRR_A = 100\%$  and  $IRR_B = 50\%$ . However, project B is a better one.

Note.  $\sum \frac{kC_t}{(1+r)^t} = 0$ . IRR remains the same if you multiply by a constant.

---

**Lecture 3, 2025/09/10**

---

Here are some ways to deal with the scaling problem:

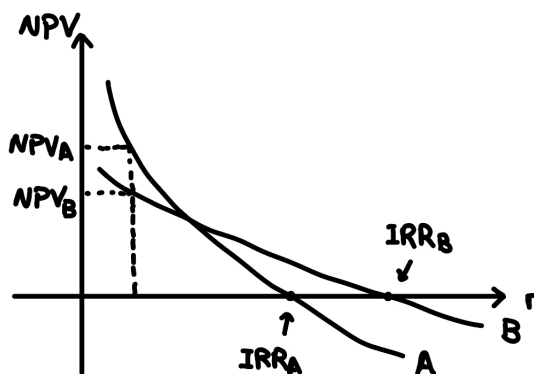
- Calculate the NPV and choose the higher one. In the example,  $IRR_A > IRR_B$ , but  $NPV_A = 1 < NPV_B$ .
- If the IRR picks the “smaller” project, then we do **incremental analysis**.

**Example.** Incremental project:  $C = B - A$  with  $C_0 = -99, C_1 = 148$ .

If the IRR of the incremental project  $< r \Rightarrow$  stick with the smaller project. If  $> r \Rightarrow$  go with the bigger project.

(4) Timing problem.

Consider pure investment projects.



According to the IRR rule, B would be chosen, but A is better because  $NPV_A > NPV_B$ .

**Example.**

Project short:  $C_0 = -300, C_1 = 345, C_t = 0$  for  $t \geq 2$ .

Project long:  $C_0 = -200, C_4 = 330, C_t = 0$  for  $t \neq 0, 4$ .

Note that

$$IRR_{\text{short}} = 15\%, \quad NPV_{\text{short}} = 13.64$$

$$IRR_{\text{long}} = 13.34\%, \quad NPV_{\text{long}} = 25.39.$$

*Note (Recall from ACTSC 231: YTM of a Bond).*

A bond is a loan in exchange for regular coupon payments and the redemption value at maturity.

We have

$$P = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}$$
$$0 = -P + \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}.$$

Note that  $r$  is the interest rate that equates NPV to 0. So YTM = IRR of the bond.

**Example.** Let  $C_0 = -100, C_1 = 10, C_2 = 110$  with  $r = 10\%$ . We can think of this as a bond with  $P = 100, C = 10, F = 110, N = 2$ . The IRR is 10%.

Also note that if the money was put in a bank, then we would have  $C_0 = -100, C_2 = 121$ .

The extra \$1 is due to the \$10 invested at 10% for another year.

IRR assumes that all cash flows can be reinvested at the IRR itself.

A more realistic assumption that cash flows can be reinvested at  $r$ , not IRR.

**Definition 1.17 (Profitability Index (PI)).**

The **profitability index** is

$$PI = \frac{\text{PV of future cash flows}}{\text{Initial Investment}} = \frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0}.$$

**Definition 1.18 (PI Rule).**

- For independent projects, choose all projects with  $PI > 1$ .
- For mutually exclusive projects, choose the project with the highest  $PI$  with  $PI > 1$ .

**Proposition 1.2.** For independent projects, PI agrees with the NPV rule.

*Proof.* Note that

$$\frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0} > 1 \implies NPV > 0.$$

□

**Disadvantages of PI rule:**

For mutually exclusive projects:

- (1) Scaling problem (same as the IRR rule).

**Example.** Let  $r = 10\%$ .

Project A:  $C_0 = -10, C_1 = 15, C_2 = 40$ .

Project B:  $C_0 = -20, C_1 = 70, C_2 = 10$ .

Note that

$$PI_A = 4.67, \quad NPV_A = 36.7$$

$$PI_B = 3.60, \quad NPV_B = 51.9.$$

Note that  $PI_A > PI_B$ , but  $NPV_A < NPV_B$ . Let's consider the incremental project  $C = B - A$  with  $C_0 = -10, C_1 = 55, C_2 = -30$ .

If the incremental project has  $PI > 1$ , then choose the bigger project; otherwise, choose the smaller project.

**Summary of Investment Rules:** NPV rule is the best.

## 1.4 Equivalent Annual Costs (EAC)

### Example (Machines with Unequal Lives).

Let  $r = 10\%$  and consider the following two machines.

A: \$500 to buy, \$120/year to operate, lasts for 3 years.

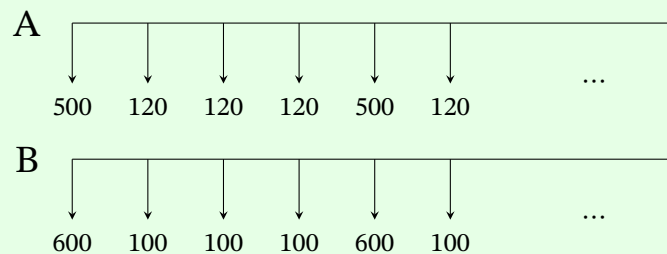
B: \$600 to buy, \$100/year to operate, lasts for 4 years.

Note that we have to use the machine (so we buy another one after it breaks down).

We can calculate the NPVs to get

$$NPV_A = -798.42, \quad NPV_B = -916.94.$$

However, the actual cash flows are different. We have the following cash flows:



Find the EAC of each machine.

*Proof.* This is equivalent to solving for  $x$  in  $NPV_A = x \cdot a_{\overline{3}|10\%}$  AND  $NPV_B = x \cdot a_{\overline{4}|10\%}$ . Then, we get

$$EAC_A = \frac{NPV_A}{a_{\overline{3}|10\%}}, \quad EAC_B = \frac{NPV_B}{a_{\overline{4}|10\%}}.$$

We choose the one with lower EAC! □

**Goal:** Calculate NPV, cash flows,  $r$ , in the right way!

## 2 Capital Budgeting

### 2.1 Rules for Calculating NPVs

Remember the following rules when calculating NPVs:

- (1) Ignore sunk costs, i.e. costs we have to incur with/without the project.

**Example.** Thaler Pizza experiment.

- (2) Include opportunity costs, i.e. cost of the next best alternative.

---

#### Lecture 4, 2025/09/15

---

- (3) Be aware of externalities, i.e. effects on other projects. One project can affect cash flows from other projects.

**Example.**

- Positive externality: synergy.
- Negative externality: erosion.

- (4) Be aware of the difference between “real” and “nominal”.

*Note.*

- Nominal Interest ( $i$ ): in dollars.
- Real interest rate ( $r$ ): amount of extra goods you can buy.
- Rate of inflation ( $\pi$ ): increase with prices.

*Example.* Suppose you have \$1 today and price of the good is \$1. If you put \$1 in the bank, it will become  $1 + i$  dollars in one year. Price of the good in one year will be  $1 + \pi$ . So the # of goods you can buy next year is

$$\frac{1 + i}{1 + \pi} = 1 + r.$$

The # of extra units of the good you can buy is (Fisher's Law):

$$r = \frac{1+i}{1+\pi} - 1 = \frac{i-\pi}{1+\pi}.$$

We can also approximate  $r$  by

$$r \approx i - \pi.$$

### Rule for Calculating Cash Flows

- Be consistent. If we are using real cash flows, use real interest rates, otherwise use nominal cash flows and nominal interest rates.

*Note.* Nominal cash flows to real cash flows:

Divide by  $(1 + \pi)^t$ .

Real cash flows to nominal cash flows:

Multiply by  $(1 + \pi)^t$ .

**Example.** Suppose all cash flows are nominal. Let  $C_0 = -1000$ ,  $C_1 = 600$ ,  $C_2 = 650$ ,  $i = 14\%$ ,  $\pi = 5\%$ . Calculate the NPV.

*Proof.*

**Method 1:** Use nominal.

$$NPV = -1000 + \frac{600}{1 + 0.14} + \frac{650}{(1 + 0.14)^2}.$$

**Method 2:** Find the real interest rate from Fisher's Law:

$$1 + r = \frac{1 + i}{1 + \pi}.$$



Let  $C'_0$  be the real cash flow at time 0, then

$$\begin{aligned}C'_0 &= C_0 \\C'_1 &= \frac{C_1}{1 + \pi} \\C'_2 &= \frac{C_2}{(1 + \pi)^2}.\end{aligned}$$

Then,

$$NPV = C'_0 + \frac{C'_1}{1 + r} + \frac{C'_2}{(1 + r)^2}.$$

We should get the same answer using both methods. □

*Remark (Why does this work?).*

Suppose all cash flows are nominal. Then,

$$\begin{aligned}NPV &= C_0 + \frac{C_1}{1 + i} + \frac{C_2}{(1 + i)^2} + \dots \\&= C_0 + \frac{\frac{C_1}{1 + \pi}}{\frac{1 + \pi}{1 + i}} + \frac{\frac{C_2}{(1 + \pi)^2}}{\left(\frac{1 + i}{1 + \pi}\right)^2} + \dots \\&= C'_0 + \frac{C'_1}{1 + r} + \frac{C'_2}{(1 + r)^2} + \dots.\end{aligned}$$

- (5) Be aware of depreciation for cash flows from accounting statements.

**Example.**

| Revenue | Cost | Depreciation | Taxes |
|---------|------|--------------|-------|
| 1500    | 700  | 600          | 34%   |

Depreciation is a tax-deductible expense! For the taxable income,

$$\begin{aligned}\text{Taxable Income} &= \text{Revenue} - \text{Cost} - \text{Depreciation} \\&= 1500 - 700 - 600 = 200.\end{aligned}$$

Also,

$$\text{Tax paid} = 0.34 \times 200 = 68.$$

Then,

$$\text{Cash Flow} = \text{Revenue} - \text{Cost} - \text{Tax paid} = 1500 - 700 - 68 = 732.$$

General approach: Let the cash flow be  $CF_t$ . Then,

$$\begin{aligned} CF_t &= \text{Revenues} - \text{Costs} - \text{Taxes} \\ &= (R_t - \text{Cost}_t) - (R_t - \text{Cost}_t - D)T_c \\ &= (R_t - \text{Cost}_t)(1 - T_c) + DT_c. \end{aligned}$$

The term  $(R_t - \text{Cost}_t)(1 - T_c)$  is called the **after tax operating cash flow (ATOCF)**.

Note that the term  $DT_c$  is the tax break we get because we can subtract depreciation.

And  $DT_c$  is the **capital cost allowance tax shield (CCATS)**.

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \dots + \frac{C_t}{(1+r)^t} \\ &= C_0 + \frac{C_1}{1+r} + \dots + \frac{(ATOCF + CCATS)}{(1+r)^t} \\ &= C_0 + PVATOCF + PVCCATS. \end{aligned}$$

Note that  $PV(ATOCF)$  is the PV without subtracting depreciation, and  $PV(CCATS)$  is the PV of all tax breaks.

*Note.* To see the above, we have

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \dots + \frac{C_t}{(1+r)^t} + \dots \\ &= C_0 + \frac{(ATOCF)_1 + (CCATS)_1}{1+r} + \dots + \frac{(ATOCF)_t + (CCATS)_t}{(1+r)^t} + \dots \\ &= C_0 + \sum_{t=1}^{\infty} \frac{(ATOCF)_t}{(1+r)^t} + \sum_{t=1}^{\infty} \frac{(CCATS)_t}{(1+r)^t}. \end{aligned}$$

**Question:** How is depreciation ( $D$ ) calculated?

**Example.** Suppose  $d = 20\%$ . A machine costs \$1 million dollars.

| Year | UCC Beginning | Depreciation ( $D$ ) | UCC End   |
|------|---------------|----------------------|-----------|
| 1    | \$1,000,000   | \$200,000            | \$800,000 |
| 2    | \$800,000     | \$160,000            | \$640,000 |
| 3    | \$640,000     | \$128,000            | \$512,000 |

Note that  $D = d \cdot (\text{UCC Beginning})$ , where UCC is the **undepreciated capital cost**.

**Half-Year Rule:** if you buy an asset, only 50% of the purchase price can be added to the first year. The remaining 50% can be added in the second year.

**Example (Half-Year Rule).**

Suppose  $d = 20\%$  and a machine costs \$1 million dollars.

| Year | UCC Beginning | Depreciation ( $D$ ) | UCC End   |
|------|---------------|----------------------|-----------|
| 1    | \$500,000     | \$100,000            | \$400,000 |
| 2    | \$900,000     | \$180,000            | \$720,000 |
| 3    | \$720,000     | \$144,000            | \$576,000 |

**Theorem 2.1 (PVCCATS Formula).**

We have

$$PV(\text{CCATS}) = \frac{C \cdot d \cdot T_c}{r + d} \cdot \frac{1 + 0.5r}{1 + r} - \frac{S \cdot d \cdot T_c}{r + d} \cdot \frac{1}{(1 + r)^n}$$

where

- $C$  = cost of the asset.
- $d$  = rate of depreciation.
- $n$  = lifetime of the asset.
- $S$  = salvage value of the asset.

## 2.2 Capital Budgeting Under Uncertainty

**Example.** MATHSOC is thinking of installing a solar panel:

- Cost: \$1 million.
- Annual operating cost: 10K.
- Lifetime: 15 years.
- Salvage: 100K.
- Revenues: \$100/hour.
- # of sunny hours: 1500.
- $r = 10\%$ .
- No taxes.

Find the NPV.

*Proof.* We have

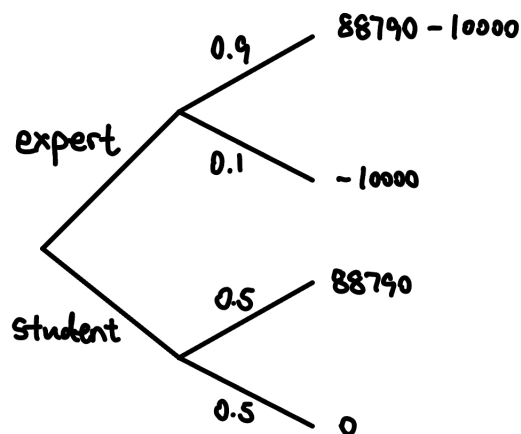
$$NPV = -1000000 - 10000a_{\overline{15}|10\%} + 150000a_{\overline{15}|10\%} + \frac{100000}{(1.1)^{15}} = 88790.$$

□

### Decision Tree

For the MATHSOC example, installation may not work. Consider two options:

- Hire an expert: cost \$10000, success rate 90%.
- Hire a student: cost \$0, success rate 50%.



There are two types of nodes in a decision tree:

- Probability nodes: calculate expected values.
- Decision nodes: choose the “better” path.

**Example (Continued).**

At the expert node, the expected value is  $0.9(88790) + 0.1(-10000) = 76911$ .

At the student node, the expected value is  $0.5(88790) + 0.5(-10000) = 39395$ .

Since  $76911 > 39395$ , we should hire the expert.

**Scenario Analysis**

For the MATHSOC example, what if the # of sunny hours  $\neq 1500$ ? Suppose # of sunny hours are 1100, 1500, 1900 with probability  $\frac{1}{3}$  each.

We calculate the expected NPV, i.e.  $\mathbb{E}[NPV] = \frac{1}{3}(NPV_{1100} + NPV_{1500} + NPV_{1900})$ .

**Monte Carlo Simulation**

Suppose sunny hours  $\sim N(1500, 200)$ . We simulate  $n$  observations from this distribution, and calculate NPVs for each outcome, and look at the distribution of NPVs.

**Sensitivity Analysis**

How sensitive is our NPV to different variables? Let  $S = \#$  of sunny hours. Then, we want to find  $\frac{dNPV}{dS} \implies$  how accurate we need to be with our estimates.

**Break-Even Analysis**

Break-even point is the value of the variable that makes  $NPV = 0$ .

**Definition 2.1 (Contribution Margin).**

The **contribution margin** is the increase in net income per unit increase in sales, i.e.

$$\text{Contribution Margin} = (\text{Unit Revenue} - \text{Unit Cost}) \cdot (1 - T_c)$$

where  $T_c$  is the corporate tax rate.

**Example.** The Tim Hortons at SLC sells donuts. Assume no taxes.

- Donuts: \$1.20.
- Cost of donuts: \$0.70.
- Fixed cost: \$100,000.

**Contribution Margin** =  $1.20 - 0.70 = 0.50$ . Then, the break-even point for # of donuts sold is  $\frac{100000}{0.50} = 200000$ , i.e. we need to sell 200000 donuts to cover the fixed cost.

**Example.** Same setting as above example, but with tax = 40%. Then,

$$\text{Contribution Margin} = (1.20 - 0.70)(1 - 0.4) = 0.30.$$

Then, the break-even point for # of donuts sold is  $\frac{100000}{0.30} \approx 333333$ .

**Example.** The IRR is the break-even interest rate!

**Example.** The discounted payback period is the break-even time!

For the next few lectures, we will focus on the following question.

**Question:** What is the right “ $r$ ” for a project? (CAPM)

## 2.3 Utility Theory

Before we get there, let’s consider the following question.

**Question:** How do rational human beings evaluate uncertain outcomes?

**Definition 2.2 (Lottery).**

A lottery  $X$  is a random variable whose outcomes are dollars.

**Example.** Consider the lotteries  $X$  and  $Y$  with the following PMFs:

|                     |     |     |
|---------------------|-----|-----|
| $X$                 | 100 | 25  |
| $\mathbb{P}(X = x)$ | 0.5 | 0.5 |

|                     |     |     |
|---------------------|-----|-----|
| $Y$                 | 81  | 49  |
| $\mathbb{P}(Y = y)$ | 0.5 | 0.5 |

One way to compare lotteries is to compare their expected values.

**Example (ST. Petersburg Paradox).**

Let  $X$  be the amount of money you win from the following lottery. A fair coin is tossed until it comes up heads. If the first head appears on the  $n$ -th toss, you win  $2^n$  dollars.

| $X$      | Probability    |
|----------|----------------|
| 2        | $\frac{1}{2}$  |
| 4        | $\frac{1}{4}$  |
| 8        | $\frac{1}{8}$  |
| 16       | $\frac{1}{16}$ |
| $\vdots$ | $\vdots$       |

The expected value of the lottery is

$$\mathbb{E}[X] = \sum x_i p_i = \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 + \frac{1}{8} \cdot 8 + \dots = 1 + 1 + 1 + \dots = \infty.$$

So you should be willing to pay any finite amount to play this game. However, in reality, most people are not willing to pay a lot of money to play this game.

**John-von-Neumann: Expected Utility Theory**

We use the preference relation  $\succeq$  to denote “is at least as good as”. That is,

$$x \succeq y \implies x \text{ is at least as good as } y.$$

**Question:** What assumptions should  $\succeq$  follow for rational agents?

**Assumptions**

- (1) Completeness:  $\forall X, Y$ , either  $X \succeq Y$  or  $Y \succeq X$  (or both).
- (2) Transitivity:  $\forall X, Y, Z$ , if  $X \succeq Y$  and  $Y \succeq Z$ , then  $X \succeq Z$ .
- (3) Independence:  $\forall X, Y, Z$  and  $\forall p \in (0, 1)$ ,  $X \succeq Y \iff pX + (1 - p)Z \succeq pY + (1 - p)Z$ .

**Theorem 2.2.** If  $\succeq$  satisfies the above three assumptions, then  $\exists$  a function  $u(\cdot)$  such that  $\forall X, Y$ ,

$$X \succeq Y \iff \mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)].$$

Note.  $u(\cdot)$  is called the **von-Neumann Morgenstern utility function**. It is the utility you get after consuming  $x$  dollars.

**Example.** Consider the following lotteries with  $u(x) = \sqrt{x}$ .

|                     |     |     |                     |     |     |
|---------------------|-----|-----|---------------------|-----|-----|
| $X$                 | 100 | 25  | $Y$                 | 81  | 49  |
| $\mathbb{P}(X = x)$ | 0.5 | 0.5 | $\mathbb{P}(Y = y)$ | 0.5 | 0.5 |

For lottery  $X$ ,

$$\mathbb{E}[u(X)] = 0.5\sqrt{100} + 0.5\sqrt{25} = 5 + 2.5 = 7.5.$$

For lottery  $Y$ ,

$$\mathbb{E}[u(Y)] = 0.5\sqrt{81} + 0.5\sqrt{49} = 4.5 + 3.5 = 8.$$

## 2.4 Risk Aversion

**Definition 2.3 (Risk Averse).**

A person is a **risk averse** if (for any lottery), they prefer the expected values for sure over the lottery itself.

**Definition 2.4 (Risk Neutral).**

A person is **risk neutral** if they only care about the expected values.

**Definition 2.5 (Risk Loving).**

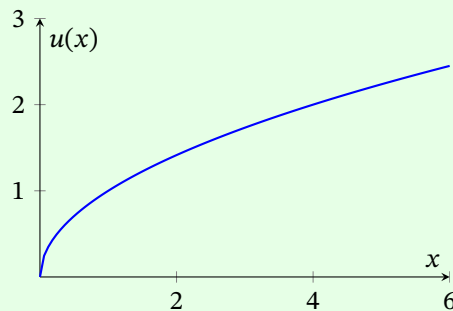
A person is **risk loving** if they prefer the lottery over the expected values.

**Proposition 2.3.** If the von-Neumann Morgenstern utility function  $u(\cdot)$

- is concave, then the person is risk averse.
- is linear, then the person is risk neutral.
- is convex, then the person is risk loving.



**Example.** Let  $u(x) = \sqrt{x}$ . Note that  $u''(x) < 0$ , so the person is risk averse.



Decreasing marginal utility over money  $\implies$  Risk averse, because the more money you have, the less utility you get from an extra dollar.

Say we have lotteries  $X$  and  $Y$  with the following PMFs:

|                     |     |     |
|---------------------|-----|-----|
| $X$                 | 100 | 0   |
| $\mathbb{P}(X = x)$ | 0.5 | 0.5 |

|                     |    |
|---------------------|----|
| $Y$                 | 50 |
| $\mathbb{P}(Y = y)$ | 1  |

Then,

$$\mathbb{E}[u(X)] = 0.5\sqrt{100} + 0.5\sqrt{0} = 5$$

$$\mathbb{E}[u(Y)] = 1 \cdot \sqrt{50} \approx 7.07.$$

If  $u(x) = x$ , then  $\mathbb{E}[u(X)] = 0.5 \cdot 100 + 0.5 \cdot 0 = 50$  and  $\mathbb{E}[u(Y)] = 1 \cdot 50 = 50$ . So the person is risk neutral.

If  $u(x) = x^2$ , then  $\mathbb{E}[u(X)] = 0.5 \cdot 100^2 + 0.5 \cdot 0^2 = 5000$  and  $\mathbb{E}[u(Y)] = 1 \cdot 50^2 = 2500$ . So the person is risk loving.

---

### Lecture 6, 2025/09/22

---

Recall that a lottery  $X$  is a random variable with dollars as outcomes, where  $x$  = outcome of  $X$ . The PMF of  $X$  is  $f(x) = \mathbb{P}(X = x)$ .

Let  $U(X)$  be the utility you get from lottery  $X$ . Recall Theorem 2.2:

If a human being is rational,  $\exists u(\cdot)$  over money such that

$$X \succeq Y \iff \mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)]$$

where  $U(X)$  = utility one gets from lottery  $X = \mathbb{E}[u(X)]$ .

**Proposition 2.4.**

- (1) A person is risk averse if  $u(\mathbb{E}[X]) > \mathbb{E}[u(X)]$  for all  $X$ .
- (2) A person is risk neutral if  $u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$  for all  $X$ .
- (3) A person is risk loving if  $u(\mathbb{E}[X]) < \mathbb{E}[u(X)]$  for all  $X$ .

**Definition 2.6 (Certainty Equivalent).**

The **certainty equivalent** of a lottery  $X$  is  $CE(X)$  = dollar value (for sure) that makes you equally happy as the lottery, i.e.

$$u(CE(X)) = \mathbb{E}[u(X)].$$

**Example.** Let  $X$  be a lottery with the following PMF:

|                     |     |     |     |
|---------------------|-----|-----|-----|
| $X$                 | 125 | 64  | 0   |
| $\mathbb{P}(X = x)$ | 1/3 | 1/3 | 1/3 |

Let  $u(x) = x^{\frac{1}{3}}$ . Then,

$$\mathbb{E}[u(X)] = \frac{1}{3}125^{\frac{1}{3}} + \frac{1}{3}64^{\frac{1}{3}} + \frac{1}{3}0^{\frac{1}{3}} = 3.$$

Then, the certainty equivalent is a solution to  $u(x) = 3$ , i.e.

$$x^{\frac{1}{3}} = 3 \implies x = 27.$$

**Proposition 2.5.**

- (1) Risk averse:  $CE(X) < \mathbb{E}[X]$ .
- (2) Risk neutral:  $CE(X) = \mathbb{E}[X]$ .
- (3) Risk loving:  $CE(X) > \mathbb{E}[X]$ .

## Degree of Risk Aversion

Note that  $u''$  measures how risk averse a person is. We use the following measures.

### Definition 2.7 (Absolute Risk Aversion (ARA)).

The **absolute risk aversion (ARA)** is

$$\text{ARA} = -\frac{u''(x)}{u'(x)}.$$

This is also called the Arrow-Pratt measure of absolute risk aversion.

### Definition 2.8 (Relative Coefficient of Risk Aversion (RRA)).

The **relative coefficient of risk aversion (RRA)** is

$$\text{RRA} = -\frac{xu''(x)}{u'(x)} = x \cdot \text{ARA}.$$

**Example.** Let  $u(x) = -e^{-\alpha x}$ . Calculate the ARA.

*Proof.* Note that

$$u'(x) = \alpha e^{-\alpha x}, \quad u''(x) = -\alpha^2 e^{-\alpha x}.$$

Then,

$$\text{ARA} = -\frac{u''(x)}{u'(x)} = -\frac{-\alpha^2 e^{-\alpha x}}{\alpha e^{-\alpha x}} = \alpha.$$

□

**Example.** Let  $u(x) = x^\alpha$  for  $0 < \alpha < 1$ . Calculate the RRA.

*Proof.* Note that

$$u'(x) = \alpha x^{\alpha-1}, \quad u''(x) = \alpha(\alpha-1)x^{\alpha-2}.$$

Then,

$$\text{RRA} = -\frac{xu''(x)}{u'(x)} = -\frac{x\alpha(\alpha-1)x^{\alpha-2}}{\alpha x^{\alpha-1}} = 1 - \alpha.$$

□

## 2.5 Mean-Variance Utility

This is a different approach to measure risk aversion.

**Definition 2.9 (Mean-Variance Utility).**

Let  $X$  be a lottery with mean  $\mu = \mathbb{E}[X]$  and variance  $\sigma^2 = \text{Var}(X)$ . The **mean-variance utility** is

$$U(X) = \mu - \frac{A}{2}\sigma^2$$

where  $A$  is a known constant.

*Note.* All mean-variance utility are risk averse.

In fact,  $A$  is the **coefficient of risk aversion**. The higher  $A$  is, the more risk averse the person is.

*Remark (Summary of Two Approaches).*

(1)  $U(X) = \mathbb{E}[u(X)]$ .

(2)  $U(X) = \mu - \frac{A}{2}\sigma^2$ .

**Question:** Are these two approaches related?

Suppose we have a von-Neumann Morgenstern utility function

$$u(x) = x - \alpha x^2.$$

Then,  $u'(x) = 1 - 2\alpha x$ . Note that we want  $u'(x) > 0 \implies x < \frac{1}{2\alpha}$ , i.e. we are looking at “small” gambles. Note that a von-Neumann Morgenstern agent maximizes

$$\mathbb{E}[u(X)] = \mathbb{E}[X - \alpha X^2] = \mathbb{E}[X] - \alpha \mathbb{E}[X^2].$$

Note that

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \implies \mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2.$$

Then,

$$\mathbb{E}[u(X)] = \mu - \alpha(\sigma^2 + \mu^2) = \mu - \alpha\sigma^2 - \underbrace{\alpha\mu^2}_{\approx 0} \approx \mu - \alpha\sigma^2.$$

**Example.** Suppose the von-Neumann Morgenstern utility function is

$$u(x) = -e^{-\alpha x}.$$

Let  $X \sim N(\mu, \sigma^2)$ . Then, this person is also a mean-variance utility maximizer.

*Proof.* We will do the proof in the next tutorial. □

We need to find  $\mu$  and  $\sigma^2$  of a lottery. We estimate  $\mu$  by the following:

$$\hat{\mu} = \frac{1}{n} \sum_{t=1}^n r_t,$$

where  $r_t$  is the return at time  $t$ .

### 3 Capital Asset Pricing Model (CAPM)

#### 3.1 Risk and Return

**Question:** How are the  $r_t$ 's calculated?

Let  $P_t$  be the price of an asset today. Then,  $P_{t+1}$  is the price of the asset in 1 year. Let  $D_t$  be the dividends you get this year. Then, the return is

$$r_t = \frac{(P_{t+1} + D_t) - P_t}{P_t} = \frac{P_{t+1} - P_t}{P_t} + \frac{D_t}{P_t}$$

where  $\frac{P_{t+1} - P_t}{P_t}$  is the **price yield** and  $\frac{D_t}{P_t}$  is the **dividend yield**.

##### **Example (Holding Period Return).**

Suppose the bank offers the following deal:

- 4% for year 1.
- 8% for year 2.
- 12% for year 3.

What is the average return?

*Proof.* Let the average be  $x$ . Then,

$$(1 + x)^3 = (1 + 0.04)(1 + 0.08)(1 + 0.12) \implies x = \sqrt[3]{1.04 \times 1.08 \times 1.12} - 1.$$

Note that  $x$  is the geometric mean in this case (not arithmetic mean). □

#### **Measure of Volatility**

##### **Definition 3.1 (Variance and Standard Deviation of Returns).**

The variance of the returns is defined as

$$\sigma^2 = \frac{1}{n-1} \sum (r_t - \mu)^2,$$

Also,  $\sigma = \sqrt{\sigma^2}$  is the standard deviation.

## Other Measures

Let  $X$  and  $Y$  be the following two distributions of returns where  $X$  has a long left tail and  $Y$  has a long right tail. Here,  $Y$  is preferred by most investors even though both have the same variance.

We need to come up with other measures of risk which capture the asymmetry.

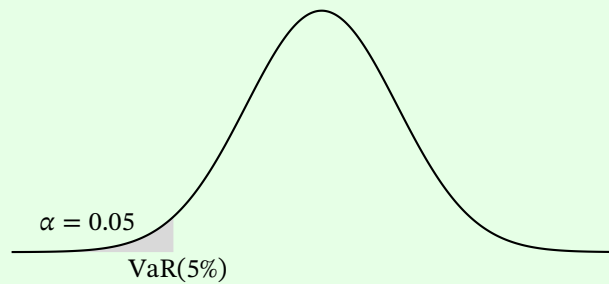
### Definition 3.2 (Value At Risk (VaR)).

$\text{VaR}(\alpha)$ , where  $\alpha$  is given as that value of  $x$  such that

$$\mathbb{P}(X \leq x) = \alpha.$$

In other words, VaR is the best case for the worst case scenarios.

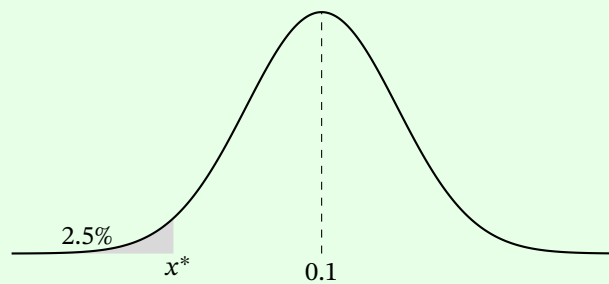
**Example.**  $\text{VaR}(5\%)$  can be interpreted in the following plot.



**Example.** Let  $X \sim N(0.1, 0.08^2)$ . Find  $\text{VaR}(2.5\%)$ .

*Proof.* Let  $x^*$  be such that  $\mathbb{P}(X \leq x^*) = 2.5\%$ . Then,

$$\mathbb{P}\left(\frac{X - \mu}{\sigma} \leq \frac{x^* - \mu}{\sigma}\right) = 2.5\% \implies \mathbb{P}(Z \leq y^*) = 0.025.$$



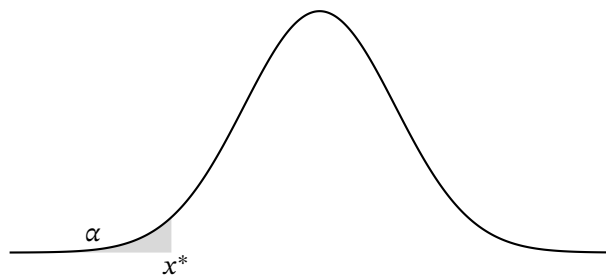
Thus,  $y^* = \frac{x^* - \mu}{\sigma} = -1.96 \implies x^* = \mu + \sigma y^* = 0.1 + 0.08 \times (-1.96) = -0.0568$ . Hence,  $\text{VaR}(2.5\%) = -5.68\%$ .  $\square$

**Definition 3.3 (Expected Loss (EL)).**

The **expected loss (EL)** or **conditional expected loss (CEL)** is defined as

$$\text{EL}(\alpha) = \text{average return you make, conditional on the fact that } X \leq \text{VaR}(\alpha).$$

Look at the following plot.



Here,  $x^* = \text{VaR}(\alpha)$  with  $\mathbb{P}(X \leq x^*) = \alpha$ . The CEL is the average of the shaded area. i.e.

$$\text{CEL}(\alpha) = \mathbb{E}[X \mid X \leq x^*].$$

---

**Lecture 7, 2025/09/24**

---

Recall some other measures of volatility:

- VaR.
- EL.
- Downside semi-variance.

**Example.** Suppose we have the following returns and probabilities.

| $X$ (%) | -15  | -10  | 0   | 10  | 20   |
|---------|------|------|-----|-----|------|
| $f(x)$  | 0.02 | 0.03 | 0.5 | 0.4 | 0.05 |



Note that  $\text{Var}(5\%) = -10\%$ . For  $\text{EL}(5\%)$ ,

| $X$ (%)           | -15 | -10 |
|-------------------|-----|-----|
| Conditional probs | 0.4 | 0.6 |

Then,  $\text{EL}(5\%) = -15\% \times 0.4 + (-10\%) \times 0.6 = -12\%$ .

**Definition 3.4 (Downside Semi-Variance).**

The **downside semi-variance** is defined as

$$\text{DSV} = \frac{1}{n-1} \sum_{r_i < \mu} (r_i - \mu)^2.$$

## 3.2 Capital Asset Pricing Model

**Question:** What portfolio will a risk-averse individual choose?

**General Problem:** Given  $n$  assets, a portfolio is  $\mathbf{w} = (w_1, \dots, w_n)$  where  $w_i$  is the proportion of wealth invested in asset  $i$ . Note that  $\sum_{i=1}^n w_i = 1$ . What should the  $w_i$ 's be?

### Capital Allocation Line

Suppose we are in a two-asset universe.

- Asset  $F$ : risk-free asset with return  $r_f$ .
- Asset  $P$ : risky asset with  $\mu_p$  = average return,  $\sigma_p$  = volatility.

**Example.** Let  $r_f = 7\%$ ,  $\mu_p = 15\%$ ,  $\sigma_p = 20\%$ . A portfolio here is  $(1 - y, y)$ , where  $y$  is the proportion invested in the risky asset and  $1 - y$  is the proportion invested in the risk-free asset. Suppose we invest \$10000. If  $y = 0.3$ , then we invest \$7000 in  $F$  and \$3000 in  $P$ .

The utility function is

$$U(x) = \mu - \frac{A}{2} \sigma^2.$$

**Question:** What should  $y$  be?

**Example.** Let  $r_f = 7\%$ ,  $\mu_P = 15\%$ ,  $\sigma_P = 20\%$ . Client 1 wants to invest with an average return of 10%. What should be  $y$ ?

*Proof.* Note that

$$(1 - y)r_f + y\mu_P = 10\% \implies y = \frac{10\% - 7\%}{15\% - 7\%} = 0.375.$$

□

Client 2 wants the volatility to be  $< 15\%$ . What should be  $y$ ?

*Proof.* Note that the portfolio of Client 2 is

$$\underbrace{\bar{C}}_Y = \underbrace{(1 - y)F}_a + y \underbrace{\bar{P}}_X.$$

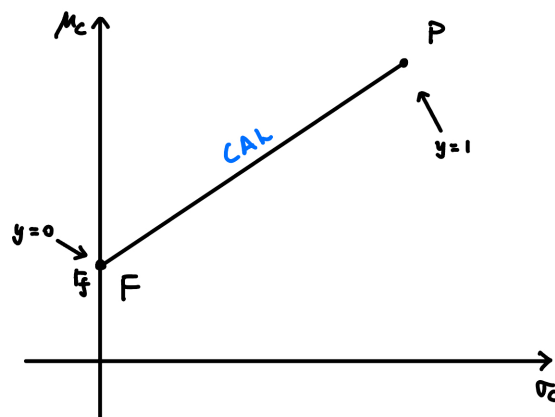
*Remark.* Recall from STAT 230, let  $Y = a + bX$ , then  $\sigma_Y = |b|\sigma_X$ .

Also,  $F$  is a constant and  $P$  is a random variable. Thus,

$$\sigma_C = y\sigma_P < 15\% \implies y < \frac{15\%}{20\%} = 0.75.$$

□

**Question:** What are the possible  $(\mu, \sigma)$  pairs we can create?



Note that  $y$  = proportion invested in the risky asset  $P$ .

**Question:** What happens if  $y$  changes?

*Proof.* Note That

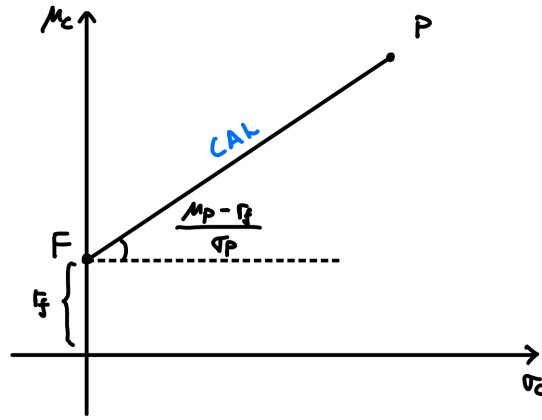
$$C = (1 - y)F + yP$$

$$\mu_C = \mathbb{E}[(1 - y)r_f + yP] = (1 - y)r_f + y\mu_P = r_f + y \underbrace{(\mu_P - r_f)}_{\text{risk premium}}.$$

Also,  $\sigma_C = y\sigma_P$ . Then,

$$y = \frac{\sigma_C}{\sigma_P} \implies \mu_C = r_f + \frac{\sigma_C}{\sigma_P}(\mu_P - r_f) = r_f + \left(\frac{\mu_P - r_f}{\sigma_P}\right)\sigma_C.$$

□



The slope of the CAL is  $\frac{\mu_P - r_f}{\sigma_P}$ , which is called the **Sharpe ratio** (or **reward-risk ratio**).

#### Definition 3.5 (Sharpe Ratio).

The ratio

$$\frac{\mu_P - r_f}{\sigma_P}$$

is called the **Sharpe ratio** or **reward-risk ratio** of the risky asset  $P$ .

#### Investor's Problem

We have the following optimization problem:

$$\max_C \mu_C - \frac{A}{2} \sigma_C^2 \text{ s.t. } C \text{ is on the CAL.}$$

To solve this,

$$\mu_C = r_f + y(\mu_P - r_f), \quad \sigma_C = y\sigma_P.$$

Then, the problem becomes

$$\max_y \{r_f + y(\mu_P - r_f)\} - \frac{A}{2}(y\sigma_P)^2.$$

Taking derivative and setting it to 0,

$$\mu_P - r_f - \frac{2Ay\sigma_P^2}{2} = 0 \implies y^* = \frac{\mu_P - r_f}{A\sigma_P^2}.$$

*Note.* This is why we have  $\frac{1}{2}$  in the utility function  $U(x) = \mu - \frac{A}{2}\sigma^2$ .

**Example.** Let  $r_f = 7\%$ ,  $\mu_P = 15\%$ ,  $\sigma_P = 20\%$ ,  $A = 4$ . What is  $y^*$ ?

*Proof.* We have

$$y^* = \frac{\mu_P - r_f}{A\sigma_P^2} = \frac{15\% - 7\%}{4 \times (20\%)^2} = 0.5.$$

□

We saw the algebraic way of looking at this portfolio problem. Now, we look at it geometrically.

## Geometric Approach

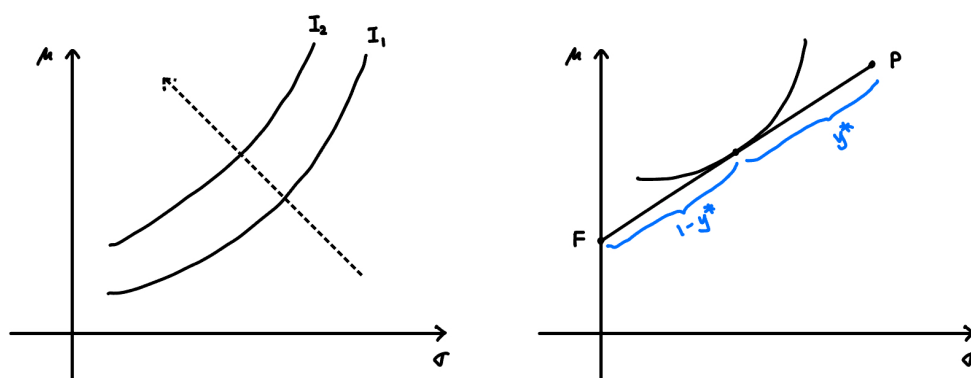
### Definition 3.6 (Indifference Curve).

An **indifference curve** is defined as

$$\{(\sigma, \mu) : \text{utility is the same for all points on the graph}\}.$$

That is, all points make you equally happy.

*Note.* Indifference curves have positive slopes.



Different indifference curves correspond to different levels of utility, and we want to move as top-left as possible. So maximizing happiness is equivalent to finding the tangent to the CAL (the most top-left indifference curve that touches the CAL).

**Question:** Can the indifference curves ever intersect?

**Answer:** No, because  $A \sim B$  and  $B \sim C$  implies  $A \sim C$ . That is, if  $A$  is as happy as  $B$  and  $B$  is as happy as  $C$ , then  $A$  is as happy as  $C$ .

*Remark.*  $X \sim Y$  means  $X$  is indifferent to  $Y$  (i.e. they provide the same utility).

**Question:** Is it possible that  $y^* > 1$ ?

**Answer:** Yes, we can borrow money and invest more in the risky asset.

## Lecture 8, 2025/09/29

**Question:** For a risk-averse, mean-variance utility agent, what portfolio will they choose?

**Case 1:** One risk-free asset  $F$  and one risky asset  $P$ .

We have a portfolio  $(1-y, y)$  where  $y$  = proportion in the risky asset. For a portfolio  $C$ , we have

$$U = \mu_C - \frac{A}{2}\sigma_C^2.$$

**Example.** Let  $r_f = 7\%$ ,  $\mu_P = 15\%$ ,  $\sigma_P = 20\%$ . Suppose  $y = 0.75$ . What are  $\mu_C$  and  $\sigma_C$ ?

*Proof.* Note that

$$\mu_C = 0.25 \cdot 7\% + 0.75 \cdot 15\% = 13\%, \quad \sigma_C = 0.75 \cdot 20\% = 15\%.$$

□

The feasible set here is a straight line called the **Capital Allocation Line (CAL)**. The equation of the CAL is

$$\mu_C = r_f + \left( \frac{\mu_P - r_f}{\sigma_P} \right) \sigma_C$$

where the slope is the **Sharpe Ratio**.

We had the optimization problem:

$$\max_C \mu_C - \frac{A}{2}\sigma_C^2 \text{ s.t. } C \text{ is on the CAL} \implies y^* = \frac{\mu_P - r_f}{A\sigma_P^2}.$$

Note that  $y^*$  depends on the risk premium  $\mu_P - r_f$ , the volatility  $\sigma_P$ , and the risk aversion  $A$ .

**Case 2:** The two risky asset model.

| Two Assets | Mean    | SD         |
|------------|---------|------------|
| $A$        | $\mu_A$ | $\sigma_A$ |
| $B$        | $\mu_B$ | $\sigma_B$ |

where  $\sigma_A, \sigma_B > 0$ . Let  $(w_A, w_B)$  be a portfolio, where  $w_A$  is the proportion invested in asset  $A$  and  $w_B$  is the proportion invested in asset  $B$ .

**Question:** What  $w_A, w_B$  should we choose?

For the feasible set, it is no longer linear (in general). We have

$$C = w_A A + w_B B, \quad w_A + w_B = 1.$$

Then,

$$\mu_C = w_A \mu_A + w_B \mu_B, \quad \sigma_C^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \text{Cov}(A, B).$$

*Remark (STAT 230 Review).*

Recall that

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X \mu_Y.$$

Let  $X = \#$  of beers drunk,  $Y = \text{ACTSC 372 grades}$ . If  $x > \mu_X$ , it is more likely that  $y < \mu_Y$ . Thus,  $\text{Cov}(X, Y) < 0$ . The idea is that  $\text{Cov}(X, Y)$  is the average of the product of the deviations from the mean. It is also the average of the product minus the product of the averages.

**Proposition 3.1 (Properties of Covariance).**

- (1)  $\text{Cov}(X, k) = 0$  for any constant  $k$ .
- (2)  $\text{Cov}(X, X) = \text{Var}(X)$ .
- (3) Bilinearity:  $\text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$  for any constants  $a, b$ .
- (4) If  $X \perp\!\!\!\perp Y$ , then  $\text{Cov}(X, Y) = 0$ . (The converse is not true in general.)

**Example (Counterexample for Converse of (4)).** Let  $X$  be such that

|        |               |               |
|--------|---------------|---------------|
| $X$    | -1            | 1             |
| $f(x)$ | $\frac{1}{2}$ | $\frac{1}{2}$ |

Let  $Y = X^2 = 1$  so that  $X$  and  $Y$  are not independent. Then,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mu_X \mu_Y = 0 - 0 \cdot 1 = 0.$$

Note that  $\text{Cov}(X, Y)$  has units. We define the following to get a unitless measure.

**Definition 3.7 (Correlation Coefficient).**

The **correlation coefficient** between two random variables  $X$  and  $Y$  is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Note that  $\rho_{XY}$  is unit independent.

**Proposition 3.2 (Properties of Correlation Coefficient).**

- (1)  $-1 \leq \rho_{XY} \leq 1$  for all  $X, Y$ .
- (2) If  $X \perp\!\!\!\perp Y$ , then  $\rho_{XY} = 0$ .
- (3) If  $\rho = 0$ , then there is no linear relationship between  $X$  and  $Y$ .

Recall that for any portfolio  $C = (w_A, w_B)$ , we have

$$\begin{aligned}\mu_C &= w_A \mu_A + w_B \mu_B \\ \sigma_C^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \text{Cov}(A, B) \\ &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B.\end{aligned}$$

**Special Case:  $\rho = 1$** 

Note that

$$\begin{aligned}\mu_C &= w_A \mu_A + w_B \mu_B \\ \sigma_C^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B = (w_A \sigma_A + w_B \sigma_B)^2.\end{aligned}$$

Then,  $\sigma_C = w_A \sigma_A + w_B \sigma_B$ . Note that  $\sigma_C$  is linear in  $w_A, w_B$ . Thus, the feasible set is a straight line.

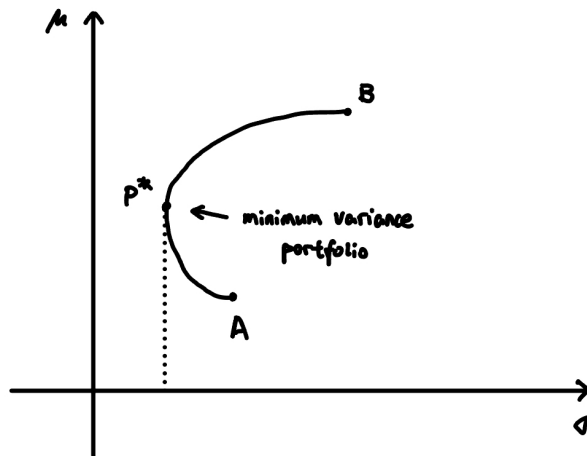
If  $\rho \neq 1$ , then  $\sigma_C < w_A \sigma_A + w_B \sigma_B$ . The feasible set will be bowed towards the top left.

**Value of Diversification**

Diversification means to invest in multiple assets. This lowers  $\sigma$  for the same  $\mu$ .

There are limits to diversification! Look at the plot below,  $p^*$  = minimum variance portfolio.





**Special Case:**  $\rho = -1$

Note that

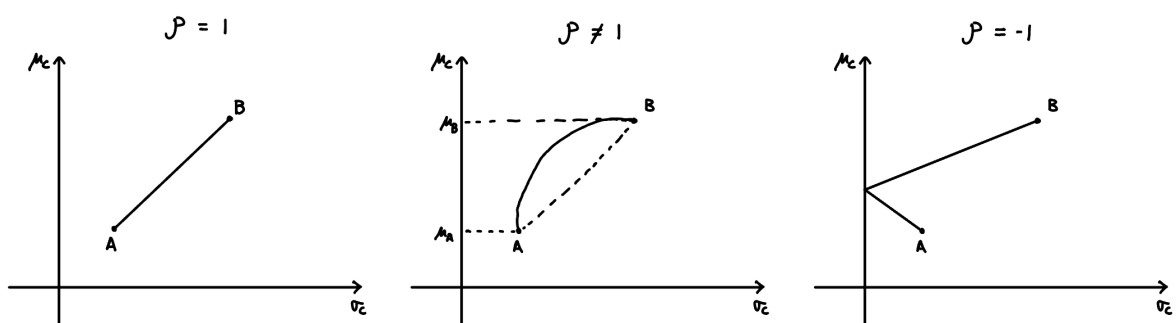
$$\begin{aligned}\sigma_C^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B (-1) \sigma_A \sigma_B \\ &= (w_A \sigma_A - w_B \sigma_B)^2 \\ \Rightarrow \sigma_C &= |w_A \sigma_A - w_B \sigma_B|.\end{aligned}$$

Choosing

$$w_A = \frac{\sigma_B}{\sigma_A + \sigma_B}, \quad w_B = \frac{\sigma_A}{\sigma_A + \sigma_B}$$

gives  $\sigma_C = 0$ . All the risk can be diversified away (i.e.  $\sigma_C = 0$ ). In general, it is not true.

Here is a summary of different  $\rho$  values:



*Note.* For  $\rho \neq 1$ , the curve is bowed towards the top left due to covariance term (lower  $\sigma_C$  for the same  $\mu_C$ , compared to  $\rho = 1$  case). For example, look at the midpoint of the line joining A and B.

**Question:** Can we find the minimum variance portfolio?

This is an optimization problem:

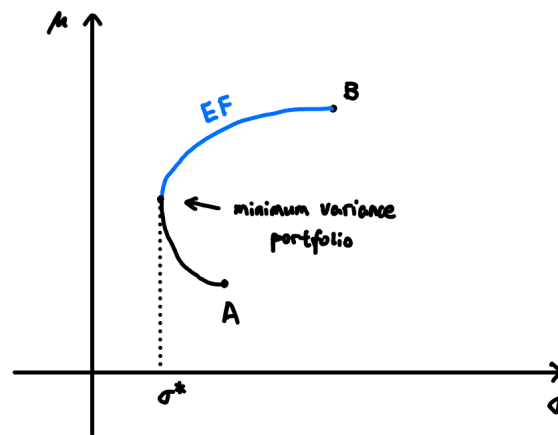
$$\min_{w_A, w_B} w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B \text{ s.t. } w_A + w_B = 1.$$

Taking derivatives and setting to 0, we get

$$w_A^* = \frac{\sigma_B^2 - \rho \sigma_A \sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho \sigma_A \sigma_B}, \quad w_B^* = 1 - w_A^*.$$

**Definition 3.8 (Efficient Frontier).**

The **Efficient Frontier** is the set of optimal portfolios, i.e. a portfolio is optimal if no other portfolio gives higher expected return AND lower risk at the same time.



All economic agents will be on the Efficient Frontier.

**Example.** Let  $\mu_A = 8\%$ ,  $\sigma_A = 12\%$ ,  $\mu_B = 14\%$ ,  $\sigma_B = 20\%$ ,  $\rho = 0.3$ . Let  $w_A = 0.8$ ,  $w_B = 0.2$ . Then,

$$\mu_C = 0.8 \cdot 8\% + 0.2 \cdot 14\% = 9.2\%.$$

Also,

$$\sigma_C = \sqrt{0.8^2 \cdot 12\%^2 + 0.2^2 \cdot 20\%^2 + 2 \cdot 0.8 \cdot 0.2 \cdot 0.3 \cdot 12\% \cdot 20\%} = 11.4\%.$$

**There are two sources of risk!**

- Systematic risk.
- Idiosyncratic risk (firm-specific risk).

---

**Lecture 9, 2025/10/01**

---

**Question:** What should be the optimum portfolio for a risk averse agent?

**Case 3:** There are  $n$  risky assets, i.e.  $A_1, A_2, \dots, A_n$ . The mean vector is

$$\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)^\top.$$

The variance-covariance matrix is

$$\Sigma_{n \times n} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix}$$

where  $\sigma_{ij} = \text{Cov}(A_i, A_j)$  and  $\sigma_{ii} = \text{Var}(A_i) = \sigma_i^2$ .

**Proposition 3.3 (Properties of  $\Sigma$ ).**

- (1)  $\Sigma$  is symmetric.
- (2) There are no zero rows in  $\Sigma$  (i.e. no risk-free assets).
- (3)  $\Sigma$  is invertible.

For  $n = 2$ , we have

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}.$$

Let  $C = (w_1, w_2, \dots, w_n)$  be a portfolio such that  $\sum_{i=1}^n w_i = 1$ , i.e.  $e^\top \mathbf{w} = 1$  where  $e = (1, 1, \dots, 1)^\top$  and  $\mathbf{w} = (w_1, w_2, \dots, w_n)^\top$ .

**Feasible Set**

**Question:** What are the different  $\mu$  and  $\sigma$  we can generate using these  $n$  assets?

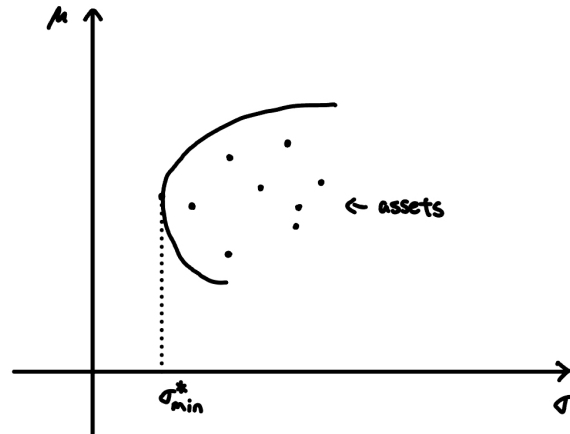
We have  $C = w_1A_1 + w_2A_2 + \dots + w_nA_n$ . Then,

$$\begin{aligned}\mu_C &= w_1\mu_1 + w_2\mu_2 + \dots + w_n\mu_n = \mathbf{w}^\top \boldsymbol{\mu}, \\ \sigma_C^2 &= \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(A_i, A_j) \\ &= w_1^2 \sigma_{11} + w_2^2 \sigma_{22} + \dots + w_n^2 \sigma_{nn} + 2 \sum_{i < j} w_i w_j \sigma_{ij} \\ &= \mathbf{w}^\top \Sigma \mathbf{w}.\end{aligned}$$

Check variance for  $n = 2$ :

$$\sigma_C^2 = \begin{pmatrix} w_1 & w_2 \end{pmatrix} \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = w_1^2 \sigma_{11} + w_2^2 \sigma_{22} + 2w_1 w_2 \sigma_{12}.$$

**Question:** What does the feasible set look like?



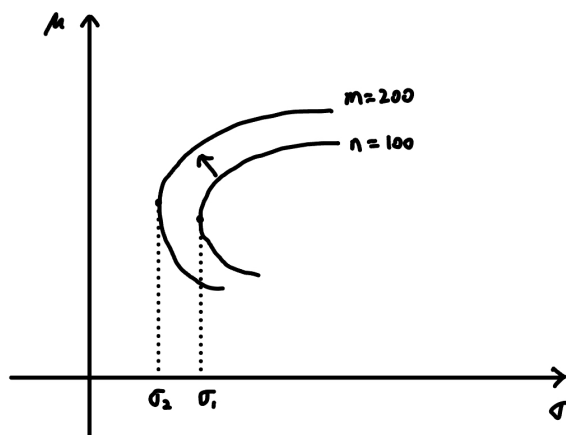
The feasible set is non-linear in general, and is bowed towards the top-left.

*Remark (Summary).*

- There is a value to diversification, i.e.  $\sigma_C^2$  can be less than the weighted average of individual variances.
- There are limits to diversification.

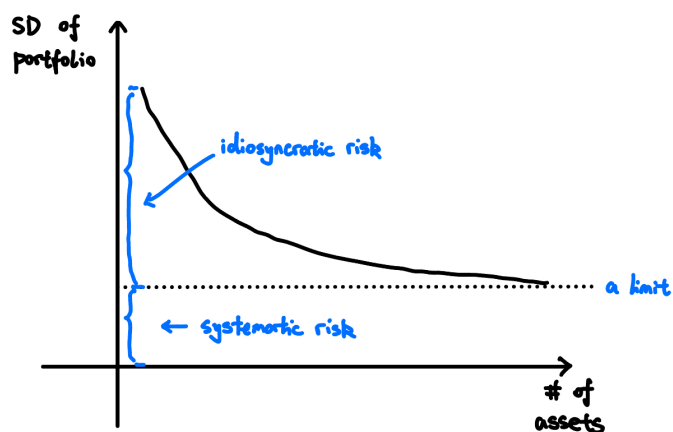
It is possible that all assets  $A_1, \dots, A_n$  lie below the feasible set.

As  $n$  increases, the feasible set moves towards the top-left:



There are two kinds of risks:

- Idiosyncratic risk (firm-specific risk).
- Systematic risk or market risk (affect all firms).



A well-diversified portfolio can get rid of all idiosyncratic risk, but not the systematic risk.

What matters are not the variances, but the covariances. That is, high-variance assets that have low covariances with other assets can reduce the overall portfolio risk, and vice versa.

**Example.** Suppose there are  $n$  assets each with the same standard deviation  $\sigma$  and the same correlation  $\rho$  (between any two assets). Consider the equally-weighted portfolio

$$\mathbf{w}^\top = \left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right) = \text{equally-weighted portfolio.}$$

What is the variance of this portfolio?

*Proof.* We have

$$\begin{aligned} \text{Var} &= \underbrace{\left(\frac{1}{n}\right)^2 \sigma^2 + \left(\frac{1}{n}\right)^2 \sigma^2 + \dots + \left(\frac{1}{n}\right)^2 \sigma^2}_{n \text{ terms}} + n(n-1) \left(\frac{1}{n}\right)^2 \rho \sigma^2 \\ &= \frac{\sigma^2}{n} + \frac{n(n-1)}{n^2} \rho \sigma^2 \\ &= \frac{\sigma^2}{n} + \frac{n-1}{n} \rho \sigma^2. \end{aligned} \quad \square$$

Note that as  $n \rightarrow \infty$ , we have  $\frac{\sigma^2}{n} \rightarrow 0$ , whereas  $\frac{n-1}{n} \rho \sigma^2 \not\rightarrow 0$ . So, covariance effect will remain.

### Some Technical Points

Here are some questions:

- (1) Find the minimum variance portfolio for  $n$ -asset problem.

Note that for  $(w_1, w_2, \dots, w_n)$ , the optimization problem is

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \text{ s.t. } \mathbf{e}^\top \mathbf{w} = 1.$$

We can use Lagrange multipliers to solve this and get

$$\mathbf{w}^* = \frac{\Sigma^{-1} \mathbf{e}}{\mathbf{e}^\top \Sigma^{-1} \mathbf{e}}.$$

- (2) How to find the Efficient Frontier (EF)?

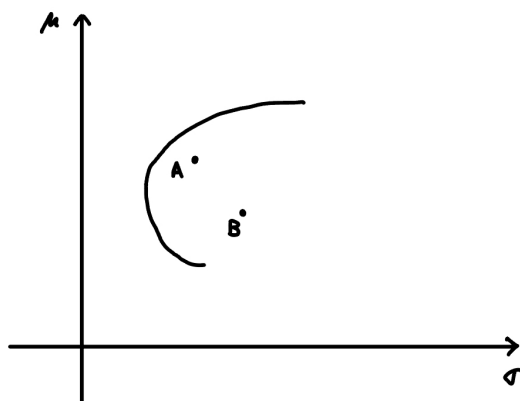
**Method 1:** Fix  $\sigma$  and maximize  $\mu$ . That is,

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu} \text{ s.t. } \mathbf{w}^\top \Sigma \mathbf{w} = \sigma^2, \quad \mathbf{e}^\top \mathbf{w} = 1.$$

**Method 2:** Fix  $\mu$  and minimize  $\sigma$ . That is,

$$\min_{\mathbf{w}} \mathbf{w}^T \Sigma \mathbf{w} \text{ s.t. } \mathbf{w}^T \boldsymbol{\mu} = \mu, \quad \mathbf{e}^T \mathbf{w} = 1.$$

Even if asset  $A$  dominates asset  $B$  (i.e.  $\mu_A > \mu_B$  and  $\sigma_A < \sigma_B$ ), it is possible to put positive weights on  $B$  because of covariances.



**Case 4:** The  $n + 1$  asset model.

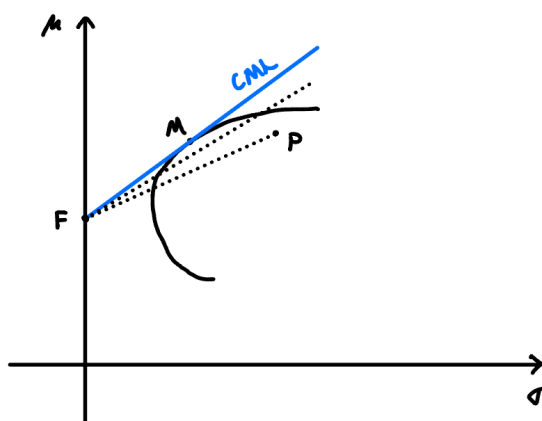
There are  $n$  risky assets  $A_1, A_2, \dots, A_n$  and one risk-free asset  $F$ .

**Question:** What should be the optimum portfolio?

**Answer:** First, solve the  $n$  risky asset problem, i.e. find the EF for the  $n$  risky assets.

**Sub-Question:** What risky asset should we choose?

**Answer:** The tangency portfolio  $M$  on the steepest CAL (i.e. CML).



All agents (irrespective of their risk aversion) will choose  $M$  as their optimum risky portfolio.

**Definition 3.9 (Capital Market Line (CML)).**

The **Capital Market Line (CML)** is the steepest CAL. It also maximizes the Sharpe ratio.

*Note.* CML is the new efficient frontier.

**Question:** What is the tangency portfolio  $M$ ?

**Example.** Let  $n = 2$  and let  $A$  and  $B$  be the two risky assets. Suppose  $M = (0.2, 0.8)$ , i.e.  $w_A = 0.2$ ,  $w_B = 0.8$ . Then, the optimum risky portfolio is 20% in asset  $A$  and 80% in asset  $B$ .

*Note.* The tangency portfolio = Market share portfolio.

**Example.** Suppose there are 3 companies with the following market shares:

- $A$ : \$5 million.
- $B$ : \$3 million.
- $C$ : \$2 million.

Then,  $M = (0.5, 0.3, 0.2)$ . The optimum risky portfolio is the market portfolio.

The optimum risky portfolio = market portfolio.

Mutual Fund Theorem states that all agents will choose the same risky portfolio  $M$  (i.e. the tangency portfolio/market portfolio) due to diversification.

ETF = Exchange Traded Fund, is a good approximation of  $M$ .

Passive investment is efficient (i.e. no need to pick stocks, just invest in the market portfolio).

---

**Lecture 10, 2025/10/06**

---

Recall:

- (1) The trade-off between a risky asset and a riskless asset is linear (CAL). Agents will choose a point on the CAL (depending on  $A$ , the coefficient of risk aversion).
- (2) The trade-off between risky assets ( $n$ -asset model) is non-linear, in general. The reason for this is the covariances.



Main points:

- Value of diversification.
- Limits to diversification (you cannot get rid of the systematic risk).

Efficiency frontier (EF): all agents (in the  $n$ -risky asset model) will be on the EF.

*Note.* On the EF, to increase  $\mu$ , you have to increase  $\sigma$ .

(3) The  $n + 1$  asset model.

$M$  = “best” risky portfolio = tangency portfolio.

CML = Capital Market Line. The slope of the CML is the Sharpe Ratio of  $M$ .

All agents will be on the CML.

Tangency Portfolio:

The tangency portfolio = market portfolio =  $(w_1, \dots, w_n)$  where  $w_i$  = market share for firm  $i$   
$$= \frac{p_i x n_i}{\sum_j \mu_j n_j}.$$

An ETF (Exchange Traded Fund) is a good approximation of  $M$ .

### 3.3 Security Market Line (SML)

**Theorem 3.4 (Capital Asset Pricing Model (CAPM)).**

Take any asset  $i$  in  $M$ . The CAPM equation is

$$\mu_i = r_f + \beta_i(\mu_M - r_f)$$

where

- $\mu_i$  = expected return of asset  $i$  (in percentage).
- $r_f$  = fixed (risk-free) rate.
- $\mu_M$  = expected return of the market.
- $\beta_i = \frac{\text{Cov}(A_i, M)}{\sigma_M^2}$  (measure of systematic risk of asset  $i$ ).

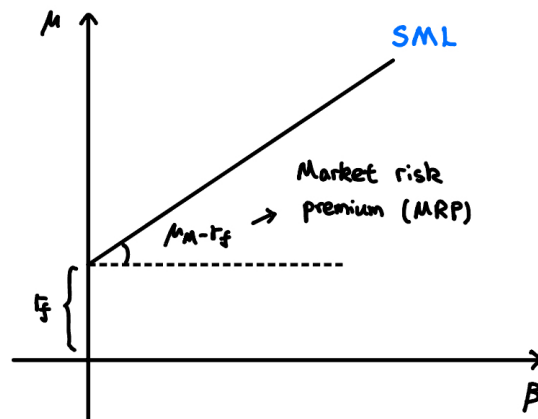
The “right” measure of risk is beta, whereas  $\sigma$  is the volatility.

*Remark.* The right interest rate for asset  $i$  is what you would have made on a similar asset (i.e. an asset with the same beta), which is given by the CAPM equation.

**Definition 3.10 (Security Market Line (SML)).**

The **Security Market Line (SML)** is the graph of the CAPM equation in the  $(\beta, \mu)$ -plane:

$$\mu_i = r_f + \beta_i(\mu_M - r_f).$$



The SML gives us the expected return of an asset, given its beta. Note that

$$\beta_i = \frac{\text{Cov}(A_i, M)}{\sigma_M^2} = \text{how sensitive is asset } i \text{ to changes in the market.}$$

- $\beta = 1 \implies$  the asset is “as sensitive” as the market.
- $\beta > 1 \implies$  the asset is “more sensitive” than the market.
- $\beta < 1 \implies$  the asset is “less sensitive” than the market.

**Example.** Consider the project:  $C_0 = -100$ ,  $C_1 = 110$ . Suppose  $r_f = 2\%$ ,  $\mu_M = 7\%$ ,  $\beta = 1.1$ . What is the “right” interest rate?

*Proof.* The average return of a “similar” project is

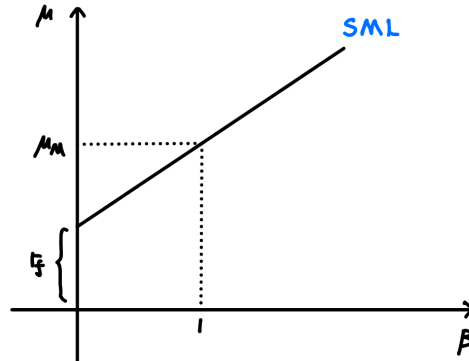
$$\mu_i = r_f + \beta_i(\mu_M - r_f) = 2\% + 1.1 \times (7\% - 2\%) = 7.5\%.$$

The interest rate for the project = expected value of a “similar” project (similar project means the same beta). □

**Question:** Suppose  $\beta = 1$ . What should be the return?

Note that

$$\mu_i = r_f + 1 \cdot (\mu_M - r_f) = \mu_M.$$



If  $\beta > 1 \implies \mu_i > \mu_M$  and if  $\beta < 1 \implies \mu_i < \mu_M$ .

If  $\beta = 0$ , then we have a risk-free asset. Note that

$$\mu_i = r_f + 0 \cdot (\mu_M - r_f) = r_f.$$

**Question:** Can we have  $\beta < 0$ ?

**Answer:** Not sure.

**Question:** Suppose asset  $A$  has  $\beta_1$  and asset  $B$  has  $\beta_2$ . Let  $C = w_1A + w_2B$ . What is  $\beta_C$ ?

*Proof.* Given that

$$\beta_A = \frac{\text{Cov}(A, M)}{\sigma_M^2}, \quad \beta_B = \frac{\text{Cov}(B, M)}{\sigma_M^2}.$$

For  $C$ , we have

$$\begin{aligned} w_1\beta_A + w_2\beta_B &= w_1 \frac{\text{Cov}(A, M)}{\sigma_M^2} + w_2 \frac{\text{Cov}(B, M)}{\sigma_M^2} \\ &= \frac{w_1 \text{Cov}(A, M) + w_2 \text{Cov}(B, M)}{\sigma_M^2} \\ &= \frac{\text{Cov}(w_1A + w_2B, M)}{\sigma_M^2} \\ &= \frac{\text{Cov}(C, M)}{\sigma_M^2} = \beta_C. \end{aligned}$$

□

**Example.**

- Beta for food: low. So expected return is low.
- Beta for cryptocurrency: high. So expected return is high.

**Question:** How to find betas?

From the SML,

$$\mu_i = r_f + \beta_i(\mu_M - r_f)$$
$$\underbrace{(\mu_i - r_f)}_{y_i} = \beta_i \underbrace{(\mu_M - r_f)}_{x_i}$$

Collect  $(x_i, y_i)$  for  $i = 1, 2, \dots, n$  and run a regression. Then,

$$\hat{\beta} = \text{the estimate for } \beta.$$

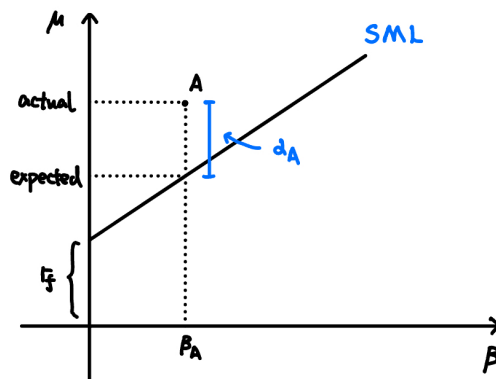
**Performance Measures**

**Definition 3.11 (Jensen's Alpha).**

The **Jensen's Alpha** of asset  $i$  is defined by

$$\alpha_i = \text{actual return of asset } i - \text{return predicted by SML}$$

- If  $\alpha > 0$ , then the asset is underpriced.
- If  $\alpha < 0$ , then the asset is overpriced.



**Definition 3.12 (Treynor Ratio).**

The **Treynor Ratio** of asset  $P$  is defined by

$$TR = \frac{\mu_P - r_f}{\beta_P}.$$

If the asset lies on the SML, then  $TR = \mu_M - r_f$ .

**Example.** XYZ is an all-equity firm with  $\beta = 1.21$ . Let  $MRP = 4\%$ ,  $r_f = 6\%$ , where MRP is the market risk premium, i.e.  $\mu_M - r_f$ . Let  $C_0 = -100$  and

| Projects | $C_1$ |
|----------|-------|
| $A$      | 140   |
| $B$      | 120   |
| $C$      | 109   |

What projects should we choose?

*Proof.* The “right” interest rate is

$$\mu_i = r_f + \beta_i(\mu_M - r_f) = 6\% + 1.21 \times 4\% = 10.84\%.$$

So  $A$  and  $B$  are profitable projects, whereas  $C$  is not. □

---

**Lecture 11, 2025/10/08**


---

Recap:

The expected return of an asset depends on the following:

- (1)  $r_f$  = risk-free rate.
- (2)  $MRP$  = market risk premium =  $\mu_M - r_f$ .
- (3)  $\beta_i$  = how sensitive your asset is to the market (appropriate measure of risk).

Note that MRP is the “common factor” for all assets and  $\beta_i$  is an appropriate measure of risk.

**Proposition 3.5 (Properties of Beta).**

- (1)  $\beta_i = 1 \implies \mu_i = \mu_M$ .
- (2)  $\beta_i > 1 \implies \mu_i > \mu_M$ .
- (3)  $\beta_i < 1 \implies \mu_i < \mu_M$ .
- (4)  $\beta_i = 0 \implies \mu_i = r_f$ .

Suppose asset  $A$  has  $\beta_1$  and asset  $B$  has  $\beta_2$ . Let  $C = w_1A + w_2B$ . Then,

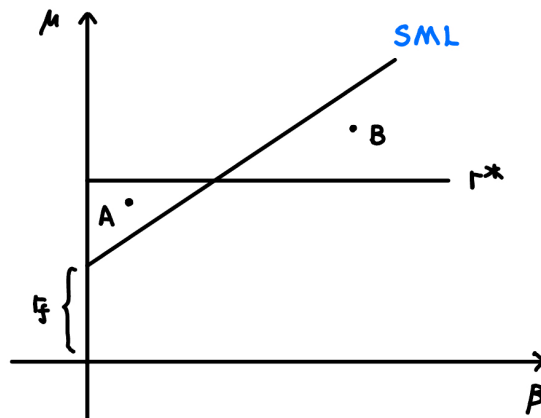
$$\beta_C = w_1\beta_A + w_2\beta_B.$$

**Definition 3.13 (Hurdle Rate).**

The **Hurdle Rate** is the minimum acceptable rate of return on an investment.

*Note.* That is, if the expected return of a project is  $> r^*$ , then accept the project; otherwise, reject.

Some companies have a fixed hurdle rate  $r^*$ , which can lead to bad investment decisions.



Note that project  $A$  is above the SML, so it is a good project. However, the hurdle rate rule suggests rejecting it. Also, project  $B$  is below the SML, but the hurdle rate rule suggests accepting it.

**Example.** Suppose a 2 risky asset model with  $\rho = 0$  and the following data:

|     | # of shares | PPS | $\mu$ | $\sigma$ |
|-----|-------------|-----|-------|----------|
| $A$ | 400         | 50  | 8%    | 12%      |
| $B$ | 1000        | 80  | 14%   | 20%      |

- (1) Find the minimum variance portfolio.

*Proof.* We have

$$w_A^* = \frac{\sigma_B^2 - \rho\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B} = \frac{20\%^2 - 0}{12\%^2 + 20\%^2 - 0} = \frac{400}{544} = 0.7353,$$

$$w_B^* = 1 - w_A^* = 0.2647.$$

So, the minimum variance portfolio is  $\begin{pmatrix} 0.7353 \\ 0.2647 \end{pmatrix}$ . □

Note that the optimization problem here is

$$\min_{w_A} w_A^2 \sigma_A^2 + (1 - w_A)^2 \sigma_B^2.$$

- (2) What is  $M$ ?

*Proof.* Company  $A$ 's value is  $400 \times 50 = 20000$ . Company  $B$ 's value is  $1000 \times 80 = 80000$ .

So,  $A$ 's market share is

$$w_A = \frac{20000}{20000 + 80000} = 0.2,$$

so the market portfolio is  $M = (0.2, 0.8)$ . □

- (3) Find  $\beta_A$ .

*Proof.* We have

$$\beta_A = \frac{\text{Cov}(A, M)}{\sigma_M^2}.$$

The numerator is

$$\text{Cov}(A, 0.2A + 0.8B) = 0.2 \text{Cov}(A, A) + 0.8 \text{Cov}(A, B) = 0.2\sigma_A^2 + 0.8 \cdot 0.$$

Plug into the formula to get  $\beta_A$ . □

- (4) Find  $\beta_B$  using an alternative method.

*Proof.* Note that  $\beta_M = 1$  and  $M = 0.2A + 0.8B$ . So,

$$\beta_M = 0.2\beta_A + 0.8\beta_B = 1 \implies \beta_B = \frac{1 - 0.2\beta_A}{0.8}.$$

We can solve for  $\beta_B$  using part (3). □

(5) Find  $r_f$ .

*Proof.* Create asset  $C$  such that  $\beta_C = 0$ :

$$C = w_A A + w_B B.$$

We must have the following:

$$\begin{cases} w_A + w_B = 1 \\ \beta_C = w_A \beta_A + w_B \beta_B = 0 \end{cases}$$

Solve for  $\hat{w}_A, \hat{w}_B$ , then

$$r_f = \mu_C = \hat{w}_A \mu_A + \hat{w}_B \mu_B.$$

□



## 4 Arbitrage Pricing Theory (APT)

### 4.1 APT and Factor Models

From CAPM,

$$\mu = r_f + \beta \cdot \text{common factor}.$$

Instead of one common factor, we allow for  $k$  common factors.

- $k = 1 \implies$  single factor model.
- $k > 1 \implies$  multi-factor model.
- $k = 1$  and the common factor is the market: market model.

#### Assumptions of APT

A1 Individual assets assumption: there are  $n$  firms, the actual return of asset  $i$  is

$$R_i = \mu_i + m_i + \epsilon_i,$$

where

- $R_i$  is the actual return of asset  $i$ .
- $\mu_i$  is the expected return of asset  $i$ .
- $m_i$  is the set of common factors.
- $\epsilon_i$  is the firm-specific individual factor (i.e. idiosyncratic error).

We have

$$m_i = \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k.$$

There are  $k$  common factors  $F_1, F_2, \dots, F_k$  with  $\beta_{ij}$  = sensitivity of  $i$ th firm to the factor  $j$ .

**Example.** The firm is Air Canada. The common factors are

$$m_i = \begin{cases} \text{Oil prices.} \\ \text{GDP of the world.} \\ \text{Inflation.} \end{cases}$$

And  $\epsilon_i$  = pilots going on strike.

For  $k = 1$ ,

$$R_i = \mu_i + \beta_i F + \epsilon_i.$$

We also assume the following:

- $\mathbb{E}[F_j] = 0 \forall j = 1, \dots, k$ , where  $F$  is the **surprise factor**.
- $\mathbb{E}[\epsilon_i] = 0$  and idiosyncratic errors are independent of each other, i.e.  $\text{Cov}(\epsilon_i, \epsilon_j) = 0$  for  $i \neq j$ .
- $\text{Cov}(F_j, F_{j'}) = 0$  for  $j \neq j'$  and  $\text{Cov}(F_j, \epsilon_i) = 0 \forall i, j$ .

A2 Portfolio assumptions: a well-diversified portfolio has  $\epsilon = 0$ .

**Example.** For  $k = 1$ , a well-diversified portfolio has

$$\begin{cases} R_1 = \mu_1 + \beta_1 F + \epsilon_1, \\ R_2 = \mu_2 + \beta_2 F + \epsilon_2, \\ \vdots \\ R_n = \mu_n + \beta_n F + \epsilon_n. \end{cases}$$

Consider the portfolio  $C$  with weights  $w_1, w_2, \dots, w_n$ :

$$C = w_1 R_1 + w_2 R_2 + \dots + w_n R_n.$$

Then,

$$R_c = \mu_c + \beta_c F + \underbrace{(w_1 \epsilon_1 + w_2 \epsilon_2 + \dots + w_n \epsilon_n)}_{=0}$$

where  $\mu_c = w_1 \mu_1 + w_2 \mu_2 + \dots + w_n \mu_n$  and  $\beta_c = w_1 \beta_1 + w_2 \beta_2 + \dots + w_n \beta_n$ .

A3 No arbitrage assumption: if two assets give you the same future returns, they must cost the same today.

**Theorem 4.1.** Under assumptions A1, A2, and A3, take any asset  $i$ ,

$$\mu_i = r_f + \beta_{i1} \gamma_1 + \dots + \beta_{ik} \gamma_k,$$

Note.  $k = 1 \implies \mu_i = r_f + \beta_i \gamma$  (i.e. CAPM). Take  $\gamma = M$ , Then

$$\mu_i = r_f + \beta_i M.$$

Since  $M = \mu_M - r_f$ , we have  $\mu_i = r_f + \beta_i(\mu_M - r_f)$ .

---

## Lecture 12, 2025/10/20

---

Recall:

- CAPM:  $\mu_i = r_f + \beta_i \cdot \text{MRP}$ , where  $\text{MRP} = \mu_M - r_f$  is the systematic factor.
- APT: instead of one systematic factor, we have multiple systematic factors, i.e. multi-factor model.
- APT Assumptions:
  - A1: Take any asset  $i$ ,

$$R_i = \mu_i + \underbrace{m_i + \epsilon_i}_{\text{noise}},$$

Noise has two components:

- \*  $m_i = \beta_{i1}F_1 + \beta_{i2}F_2 + \dots + \beta_{ik}F_k$  where  $\beta_{ij}$  = sensitivity of the  $i$ th asset to the  $j$ th factor and  $F_j$  = common factor  $j$ .

**Example.** Let  $F_j$  = oil prices,  $i$  = Ford, and  $i'$  = Tesla. Note that betas can act in the opposite direction for the same factor.

- \*  $\epsilon_i$  = idiosyncratic factor = firm-specific noise.

We assume

- \*  $\mathbb{E}[F_j] = 0 \forall j$ .
- \*  $\mathbb{E}[\epsilon_i] = 0 \forall i$  and  $\perp (\epsilon_i, \epsilon_j) \implies \text{Cov}(\epsilon_i, \epsilon_j) = 0$  for  $i \neq j$ .
- \*  $\text{Cov}(F_j, F_{j'}) = 0 \forall j \neq j'$  and  $\text{Cov}(F_j, \epsilon_i) = 0 \forall i, j$ .

- A2: Take any well-diversified portfolio  $i$ . In such a portfolio,  $\epsilon = 0$ . Thus,

$$R_i = \mu_i + m_i.$$

- A3: No arbitrage assumption. If two assets give you the same payoff, the price must be the same today.

Summary of A1 and A2:

- For individual assets:  $R_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k + \epsilon_i$ .
- For portfolios:  $R_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k$ .

### Special Cases

- $k = 1$  (single factor model) for portfolios:  $R_i = \mu_i + \beta_i F$ .
- $k = 1$  and  $F = M$  (market factor model) for portfolios:  $R_i = \mu_i + \beta_i M$ . Then,  $\mathbb{E}[R_i] = \mu_i$  and  $\text{Var}(R_i) = \beta_i^2 \sigma_M^2$ .

Recall Theorem 4.1: Under assumptions A1, A2, and A3, take any asset  $i$ ,

$$\mu_i = r_f + \beta_{i1}\gamma_1 + \cdots + \beta_{ik}\gamma_k.$$

Special case: for  $k = 1$ ,  $\mu_i = r_f + \beta_i \gamma$ .

*Proof of Theorem 4.1.*

We prove the case  $k = 1$ . Let us take two portfolios  $A$  and  $B$ . Then,

$$R_A = \mu_A + \beta_A F,$$

$$R_B = \mu_B + \beta_B F.$$

**Question:** Can we create an asset  $C$  such that  $C$  is risk-free?

We need to find weights  $w_A, w_B$  such that

$$w_A + w_B = 1, \quad w_A \beta_A + w_B \beta_B = 0.$$

Solving this, we have

$$w_A = \frac{\beta_B}{\beta_B - \beta_A}, \quad w_B = \frac{-\beta_A}{\beta_B - \beta_A}.$$

By no-arbitrage assumption, this asset  $C$  should give us  $\mu_C = r_f$ . So,

$$r_f = w_A \mu_A + w_B \mu_B.$$

Then,

$$r_f = \frac{\beta_B}{\beta_B - \beta_A} \mu_A + \frac{-\beta_A}{\beta_B - \beta_A} \mu_B \implies \frac{\mu_A - r_f}{\beta_A} = \frac{\mu_B - r_f}{\beta_B} = \gamma.$$

Thus,

$$\mu_i = r_f + \beta_i \gamma \quad \text{for } i = A, B.$$

□

### Market Model

We have

$$\mu_i = r_f + \beta_i M$$

for all portfolios. Here,

$$\mu_M = r_f + \beta_M M, \quad M = \mu_M - r_f.$$

So,  $\mu_i = r_f + \beta_i(\mu_M - r_f)$ .

**Example.** Suppose we have a two factor model. We have

$$R_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \epsilon_i.$$

Let  $A, B$ , and  $C$  be three well-diversified portfolios with the following.

| Portfolio $i$ | $\beta_{i1}$ | $\beta_{i2}$ | $\mu_i$ |
|---------------|--------------|--------------|---------|
| $A$           | 1.0          | 1.5          | 10%     |
| $B$           | 0.5          | 1.0          | 7%      |
| $C$           | 0.75         | 1.25         | 9%      |

Can we create a portfolio  $D$  using just  $A$  and  $B$  such that  $D$  is independent of factor 1?

*Proof.* We find weights  $w_A, w_B$  such that

$$w_A + w_B = 1, \quad w_A \beta_{A1} + w_B \beta_{B1} = 0.$$

□

What is the risk-free rate implied in this model?

*Proof.* Let  $w_A, w_B, w_C$  be the weights for a risk-free portfolio. We have

$$\begin{cases} w_A + w_B + w_C = 1, \\ 1.0w_A + 0.5w_B + 0.75w_C = 0, \\ 1.5w_A + 1.0w_B + 1.25w_C = 0. \end{cases}$$

We can solve for  $w_A, w_B, w_C$ . Then, the return of this portfolio (i.e. the risk-free rate) is

$$w_A\mu_A + w_B\mu_B + w_C\mu_C = r_f.$$

□

*Note.* APT is more practical than CAPM since the number of variables that need to be estimated under APT is less than that under CAPM.

## 4.2 Factors Affecting Beta

We have the following factors affecting beta:

- Cyclicity of revenues (how sensitive stocks are to business cycles).
- Operating leverage.
- Financial leverage.

### Cyclicity of Revenues

An asset that is more sensitive to business cyclese will have a higher beta.

#### **Example.**

- Groceries: low beta.
- Restaurants: higher beta.
- Luxury goods: high beta.

## Operating Leverage

**Definition 4.1 (Operating Leverage).**

**Operating leverage** is defined by

$$OL = \frac{\% \text{ change in earnings}}{\% \text{ change in sales}}.$$

Higher operating leverage  $\implies$  higher beta.

**Example.** The following information for a project is given.

- Revenue: \$250 per year forever.
- Cost: \$50 per year forever.
- Assume CAPM holds and  $r_f = 5\%$ ,  $E[r_M] = 10\%$ ,  $\beta = 0.6$ .

What is the NPV of the project?

*Proof.* We find the “right” interest rate first.

$$\mu = r_f + \beta(E[r_M] - r_f) = 0.05 + 0.6(0.10 - 0.05) = 0.08.$$

Then,

$$NPV = \frac{250}{0.08} - \frac{50}{0.08} = \$2500.$$

□

Suppose the cost was fixed. Then, what is the NPV of the project?

*Proof.* The “right” interest rate for revenues is 8%. The “right” interest rate for costs is  $r_f = 5\%$ .

Then,

$$NPV = \frac{250}{0.08} - \frac{50}{0.05} = \$2125.$$

□

**Case 1:** The project is less sensitive to sales.

**Case 2:** The project is more sensitive to sales.

**Example (Continued).**

What is the beta of the project where the cost is fixed?

*Proof.* Recall that

$$NPV = NPV_{\text{revenues}} + NPV_{\text{costs}} = \frac{250}{0.08} - \frac{50}{0.05}.$$

Then,

$$w_{\text{revenues}} = \frac{3125}{2125} = 1.47, \quad w_{\text{costs}} = \frac{-1000}{2125} = -0.47.$$

Thus,

$$\beta_{\text{project}} = 1.47 \cdot 0.6 + (-0.47) \cdot 0 = 0.88.$$

□

**Financial Leverage**

So far, we assumed all equity firms.

**Question:** What is the right interest rate if a firm has debt?

---

**Lecture 13, 2025/10/22**


---

**Information Asymmetry**

In Finance, this is called agency problems.

**Example.** In a second-hand car, suppose 50% of cars are lemons (bad cars) and 50% are good cars. A lemon is worth \$1000 and a good car is worth \$3000. Then, the buyer should only pay for \$1000 and accept the fact that they might get a bad car.

**Example.** Suppose a firm has a cash flow of \$C in perpetuity. So,

$$NPV = \frac{C}{r}$$

where  $r$  is the cost of capital.

**Liquidity**

Higher liquidity  $\implies$  easier to trade  $\implies$  lower cost of capital.



**Definition 4.2 (Bid-Ask Spread).**

The **bid-ask spread** is defined by

$$\text{Bid-Ask Spread} = \text{Buying price} - \text{Selling price}.$$

*Note.* A bid-ask spread is higher if there is information asymmetry.

**Financial Leverage**

So far, we assumed that all firms are all equity. Now, we will allow for debt.

**Question:** What is the right interest rate if a firm has debt?

**Properties of Equity Contracts**

- Ownership contract.
- Residual claim.
- Limited liability.

**Properties of Debt Contracts (Loan Contracts)**

- Interest you pay is tax-deductible.
- Failure to pay interest may lead to bankruptcy.

*Remark.* Value of firm  $V = D + E$  where  $D$  = market value of debt,  $E$  = market value of equity.

*Note.* Levered firms:  $D > 0$  and unlevered firms:  $D = 0$ .

**Questions:**

- (1) What is the optimum debt-equity ratio?

**Answer:** We will answer this later (Modigliani-Miller Theorems)

- (2) Given. the debt-equity ratio, what is the “right” interest rate?

**Answer:** Weighted Average Cost of Capital (WACC).

### 4.3 Weighted Average Cost of Capital (WACC)

**Case 1:** No taxes. The WACC is

$$r_{\text{WACC}} = \frac{E}{D+E}r_E + \frac{D}{D+E}r_D$$

where we get  $r_E$  from the SML, and  $r_D$  is the interest rate on loans (debt).

**Case 2:** With taxes. The WACC is

$$r_{\text{WACC}} = \frac{E}{D+E}r_E + \frac{D}{D+E}r_D(1 - T_c),$$

where  $T_c$  is the tax rate.

**Example.** Suppose  $D = \$1$  million,  $r_D = 8\%$ , and tax rate = 40%. Then,

$$r_D(1 - T_c) = 8\% \cdot (1 - 0.4) = 4.8\%.$$

Interest paid is \$80,000 and Tax break =  $0.4 \cdot 80,000 = \$32,000$ . So,

$$\text{Effective interest} = 80,000 - 32,000 = 48,000.$$

*Note.* Interest is tax-deductible.

**Example.** A firm has the following information.

- Debt-equity ratio is 0.6.
- Cost of debt is 15.15% (i.e.  $r_D$ ).
- Cost of equity is 20% (i.e.  $r_E$ ).
- Tax rate is 34% (i.e.  $T_c$ ).

Is this a good project?

*Proof.* The WACC is

$$r_{\text{WACC}} = \frac{D}{D+E}r_D(1 - T_c) + \frac{E}{D+E}r_E.$$

Note that  $D = 0.6E$ . Then, we can compute to get  $r_{\text{WACC}} = 16.25\%$ . Then, plug  $r_{\text{WACC}}$  (as the interest rate) into NPV formula to see if the project is good.  $\square$

**Example.** XYZ is a firm with the following information.

- Common shares: # of shares = 8.5 million, PPS = \$34,  $\beta = 1.2$ .
- Bonds: 200000 outstanding bonds ( $F = 1000$ ), 15 year bonds, 7.5% semi-annual coupons are selling for 93% of par.
- Tax rate = 35%, MRP = 7%,  $r_f = 5\%$ .

Find  $r_{WACC}$ .

*Proof.* We have

$$r_{WACC} = \frac{E}{D+E}r_E + \frac{D}{D+E}r_D(1-T_c).$$

Note that

$$E = \# \text{ of shares} \times \text{PPS} = 8.5 \text{ million} \times 34 = 289 \text{ million}.$$

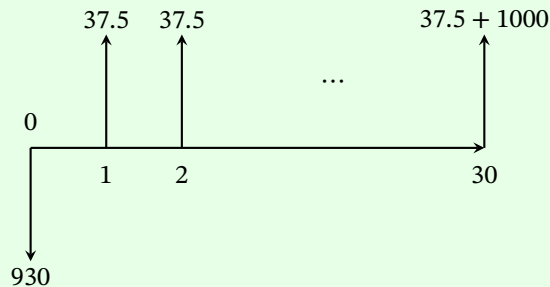
And,

$$D = \text{number of bonds} \times \text{price per bond} = 200,000 \times 930 = 186 \text{ million}.$$

To find  $r_E$ , use CAPM:

$$r_E = r_f + \beta \cdot \text{MRP} = 5\% + 1.2 \cdot 7\% = 13.4\%.$$

To find  $r_D$ , the cash flow of the bond is



Then, equate the NPV to 0 and solve for  $r_D$ :

$$0 = -930 + \sum_{t=1}^{30} \frac{37.5}{(1+r_D)^t} + \frac{1000}{(1+r_D)^{30}}.$$

Thus, we can compute to get  $r_{WACC} = 10.27\%$ . □

## 5 Efficient Market Hypothesis (EMH)

---

### Lecture 14, 2025/10/27

---

**Question:** What are the conditions for a “good” stock market?

**Sub-question:** Should the price movements be predictable or unpredictable?

**Answer:** Unpredictable. Every stock market should have this feature!

**EMH Assumptions:**

- (1) Price movements are unpredictable.
- (2) The price of a stock should reflect all relevant information about the stock.
- (3) The price adjustment should be instantaneous.

**Three Types of EMH:**

- (1) Weak form.
- (2) Semi-strong form.
- (3) Strong form.

**Definition 5.1 (Weak Form).**

The price movement of a stock tomorrow is independent of the price today. That is,

$$P_t = P_{t-1} + \epsilon_t$$

where  $\epsilon_t$  is a random variable that is independent of  $P_t$ 's.

This is called the **random walk hypothesis**.

**Example (Additional Example: Gambler's Ruin).**

Suppose person  $A$  and person  $B$  have  $\$N$  together. Say  $A$  has  $\$i$  and  $B$  has  $\$N - i$ . They play a game where at each round,  $A$  wins  $\$1$  from  $B$  with probability  $\mathbb{P}(A) = p$  and loses  $\$1$  to  $B$  with probability  $\mathbb{P}(B) = 1 - p$ . The game ends when one person loses all their money. Let  $\mathbb{P}(i)$  be the probability that  $A$  will win the game if  $A$  starts with  $\$i$ . What is  $\mathbb{P}(i)$ ?

*Proof.* Note that

$$\mathbb{P}(i) = p \cdot \mathbb{P}(i+1) + (1-p)\mathbb{P}(i-1) \quad \text{for } i = 1, 2, \dots, N-1.$$

Also,  $\mathbb{P}(0) = 0$  and  $\mathbb{P}(N) = 1$  are the boundary conditions. For example, let  $p = \frac{1}{2}$ . Then,

$$\mathbb{P}(i) = \frac{i}{N}.$$

Say  $N = 200$ ,  $p = 0.49$ ,  $i = 100$ . Then,  $\mathbb{P}(i) < 0.01$ , so don't gamble!

□

### Technical Analysts

They claim that they can see patterns in stock price movements. Then, the weak form  $\implies$  technical analysts are guessing.

#### **Definition 5.2 (Semi-Strong Form).**

The price of a stock takes into account all relevant public information.

*Note.* That is, one cannot beat the market.

### Fundamental Analysts

Under the semi-strong form, fundamental analysts are also guessing.

---

### Midterm Review

- MC/Short answer questions.
- 5 Problems,
- Total points: 44.
- Don't spend too much time.
- Don't overthink.

### **Investment Rules**

- Can we calculate NPV, IRR, PI for simple cash flow problems?
- Do we know the weaknesses of IRR, PI, DPP?
- Do we know independent and mutually exclusive projects?
- Do we know when to do incremental analysis (for IRR, PI)?
- Do we know the weaknesses of NPV? (E.g. projects with unequal lives and EAC).

### **Capital Budgeting**

- Do we know how to deal with nominal and real interest rates? (E.g. from one to another, and how to compare cash flows, i.e. be consistent).
- Can we find cash flows from accounting statements, i.e.  $(R_t - C_t)(1 - T_c) + DT_c$ ?
- Do we know what PVCCATS means?
- Do we know the half-year rule?
- Capital budgeting under uncertainty:
  - Decision Trees.
  - Sensitivity analysis, etc.

### **Utility Theory**

- What is the von-Neumann Morgenstern utility function?
- What is risk aversion, risk neutral, risk loving?
- What is certainty equivalent?
- Do we know the degrees of risk aversion?
- Do we know the relationship between the von-Neumann Morgenstern utility function and the mean-variance utility function?

### **CAPM**

- What is the CAL? That is, one risky asset + one riskless asset.
  - What means and variances can we generate?

- What is the amount to invest in the risky asset? That is,  $y^* = \frac{\mu_P - r_f}{A\sigma_P^2}$ .
- What is the sharpe ratio? That is, the performance indicator.
- Two risky assets and  $n$  risky assets models.
  - Can we construct the minimum variance portfolio?
  - $M$  = tangency portfolio = market portfolio (portfolio of market shares).
- The  $n + 1$  asset model.
  - What is the SML? That is,  $\mu = r_f + \beta(\mu_M - r_f)$ , which is the right interest rate.
  - What is beta? That is,  $\beta_A = \frac{\text{Cov}(A, M)}{\sigma_M^2}$ . Also,  $\beta_A = 1$  for market portfolio.
- Other measures of performance:
  - Jensen's alpha.
  - Treynor ratio.

#### APT

- Individual asset assumption:  $R_i = \mu_i + m_i + \epsilon_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \dots + \beta_{ik}F_k + \epsilon_i$ . What is the variance of  $R_i$ ?
- Portfolios:  $R_i = \mu_i + m_i$ .
- Suppose a two-factor model. Say

| Asset | $\beta_{i1}$ | $\beta_{i2}$ |
|-------|--------------|--------------|
| $A$   | 0.75         | -1           |
| $B$   | 1.25         | 0.25         |

Can we create portfolios that are independent of certain factors?

#### WACC

- What is the right interest rate if  $D > 0$ ? That is,  $r_{WACC} = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - T_c)$ , where  $r_E$  is from the SML and  $r_D$  is from the YTM of debt.

---

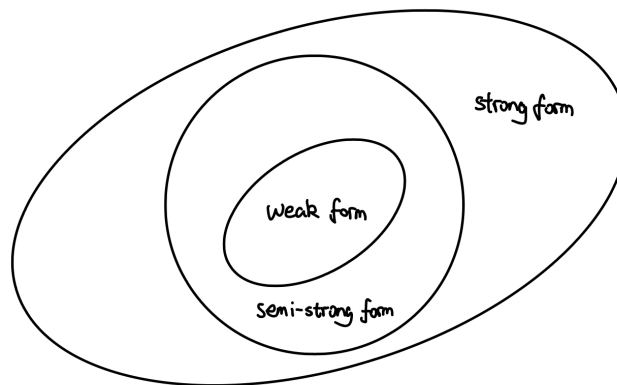
## Lecture 15, 2025/11/03

---

*Note.* Today's lecture is definition heavy and may be examined in multiple choice questions.

Recall:

- **Weak form:** you cannot “beat” the market if we know today's prices. We have  $P_t = P_{t-1} + \epsilon_t$ , e.g.  $\epsilon_t \sim N(0, \sigma^2)$ . This is called the random walk hypothesis. Sometimes the formula is written as  $P_{t+1} = P_t + g + \epsilon_t$  which is called the random walk with drift.
- **Semi-strong form:** you cannot “beat” the market with public information about the asset (e.g. stock).
- **Strong form:** you cannot “beat” the market with private information about the asset.



### Inefficiencies of the EMH

**Technical analysts:** they claim that  $P_{t+1}$  is based on  $P_t$

Weak form  $\implies$  technical analysts are guessing.

**Fundamental analysts:** they check whether an asset is over/under priced based on public information.

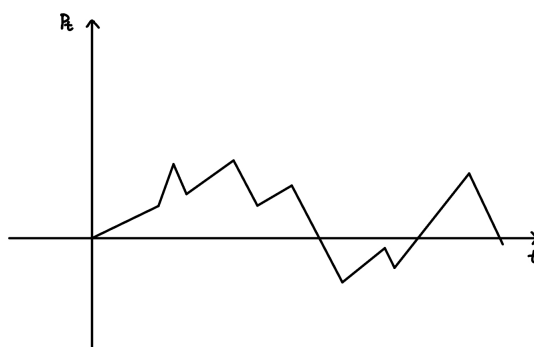
Semi-strong form  $\implies$  fundamental analysts are guessing.

Strong form  $\implies$  insider trading is not profitable, where insider trading means illegal trading based on private information.

### Evidence For/Against EMH

**Question:** Can we check whether the weak form is true?



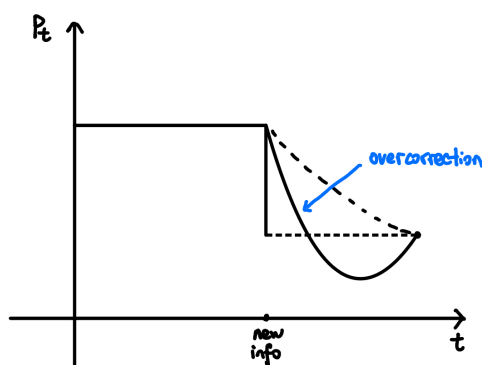


We try to “fit” a random walk to describe the stock price movements. For example, for tossing a fair coin, we can define

$$X_t = \begin{cases} 1 & \text{if H} \\ -1 & \text{if T.} \end{cases}$$

Then,  $S_n = \sum_{t=1}^n X_t$  is a random walk.

If you can find an investment manager who can consistently beat the market, then the semi-strong form is false.



**Absence of Trades:** this is an evidence for the EMH, because if all stock prices are correctly priced, then there is no incentive to trade.

**Joint Hypothesis Problem:** Testing if stocks are correctly priced is an evidence for the EMH, or an evidence for CAPM?

**Two Schools of Thought:**

- **Classical:** Dutch Book Theorem.
- **Behavioural Finance:** markets are not rational.

## 6 Capital Structure

### 6.1 The Basics

**Question:** Does leverage have any effect on the value of the firm?

**Question:** What is the “optimum” D/E ratio?

**Answer:** The Modigliani-Miller Theorem.

#### Equity

##### **Definition 6.1 (Equity).**

Equity can be one of the following:

- Ownership contract.
- Residual claims.
- Limited liability.

##### **Definition 6.2 (Market CAP).**

**Market CAP** is defined by

$$\text{Market CAP} = \text{Number of shares} \times \text{Price per share.}$$

Types of shares:

- Common shares: 1 share = 1 vote.
- Superior shares: 1 share > 1 vote.
- Limited shares: 1 share < 1 vote.

Shareholders elect the board of directors. The directors then hire management. This is how companies are run.

#### **Voting:**

- Non-cumulative voting.
- Cumulative voting.

**Example.** Suppose  $A$  has 75% of shares,  $B$  has 25% of shares, and  $B$  of directors are to be elected.

- Non-cumulative voting:  
Sequential elections:  $A$  has 75 votes and  $B$  has 25 votes. So,  $A$  will be able to select
- Cumulative voting:  $A$  will have  $75 \times 4 = 300$  votes and  $B$  will have  $25 \times 4 = 100$  votes.  
Elections are simultaneous:  $B$  can allocate all 100 votes to 1 candidate. So,  $B$  will be able to elect 1 director.

Cumulative voting  $\implies$  Minority shareholders have more power.

**Definition 6.3 (Preferred Shares).**

A **preferred share** specifies the amount of dividend we get every period.

*Note.* Preferred shares are between equity and debt.

- Like equity: not paying dividends  $\nRightarrow$  bankruptcy.
- Like debt: pre-specified payments.

We have the following types of preferred shares:

- Fixed preferred shares: the dividend rate is fixed.
- Floating preferred shares: changes with the interest rate (e.g. T-bill rates + 3%).
- Callable preferred shares: the issuer (seller) has a right to call the share back at a pre-specified price.
- Retractable preferred shares: the holder (buyer) has a right to sell the stock back to the issuer at a pre-specified price.
- Redeemable preferred shares: the buyer has the right to redeem the preferred share in exchange for common shares.
- Reset preferred shares: the dividend rate changes (typically every 5 years).

## Debt

### **Definition 6.4 (Debt).**

Features of debt:

- Loan contract.
- Interest you pay on debt is tax deductible.
- Failure to pay interest  $\implies$  bankruptcy.

Two categories of debt:

- Secured debt: backed by an asset (e.g. mortgage).
- Unsecured debt: not backed by an asset. This is also called a **debenture**.

Types of debt:

- Long term debt
- Short term debt
- Trades payable
- Indenture: the contract between the borrower and the lender.

**Example.** Restrictions on dividends, hiring good auditors.

- Subordinated debt: your debt is “behind the line” in the event of a bankruptcy.

In the event of a bankruptcy, investors get paid in the following order:

- Senior bond holder
- Junior bond holder
- Debentures
- Trades payable
- Preferred shareholders

## 6.2 Modigliani Miller Propositions

### Definition 6.5 (Levered Firm).

A **levered firm** is a firm that has debt, i.e.  $D > 0$ .

Suppose the firm produces  $\$C/\text{year}$  in perpetuity. Then, the value of the firm is

$$V = \frac{C}{r_{\text{WACC}}}.$$

Recall that

$$r_{\text{WACC}} = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - T_c).$$

**Question:** What  $D/E$  ratio should we choose to maximize the firm value (i.e. minimize  $r_{\text{WACC}}$ )?

This brings us to Modigliani and Miller Propositions.

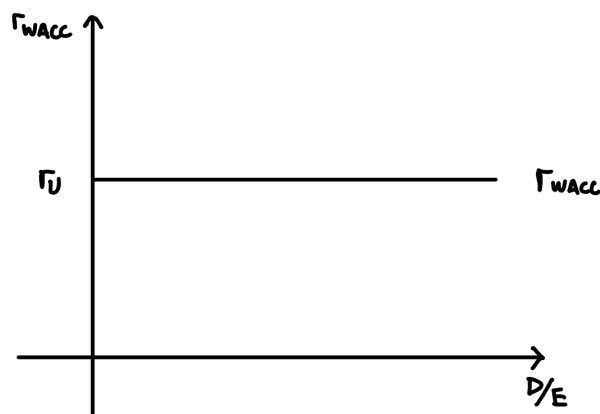
**Case 1:** No taxes, i.e.  $T_c = 0$ .

Let  $V_L$  = value of the levered firm, and  $V_U$  = value of the unlevered firm.

### Proposition 6.1 (M & M Proposition 1 (No Taxes)).

$$V_L = V_U.$$

**Implications:**  $D/E$  ratio is irrelevant to the firm value AND  $r_{\text{WACC}}$  is unaffected by the  $D/E$ .



**Example.** A firm is unlevered (i.e.  $D = 0$ ). The value of the firm is \$8000. There are 400 outstanding shares. So,

$$\text{PPS} = \frac{8000}{400} = \$20.$$

The firm is considering taking debt. They borrow \$4000 and buy off 200 shares,  $r_D = 10\%$ . What is the  $D/E$  ratio after the buyback?

*Proof.* Before leverage, i.e. unlevered:

$$D = 0, \quad E = 8000.$$

After leverage, i.e. levered:

$$D = 4000, \quad E = 8000 - 200 \times 20 = 4000 \implies \frac{D}{E} = 1.$$

Note that the value of the firm remains the same for both cases. □

**Example (No Debt Scenario).**

|          | Recession                | Normal | Expansion |
|----------|--------------------------|--------|-----------|
| Earnings | \$400                    | \$1200 | \$2000    |
| ROA      | $\frac{400}{8000} = 5\%$ | 15%    | 25%       |
| ROE      | 5%                       | 15%    | 25%       |
| EPS      | $\frac{400}{400} = \$1$  | \$3    | \$5       |

**Example (Debt Scenario).**

|                            | Recession                  | Normal                  | Expansion                |
|----------------------------|----------------------------|-------------------------|--------------------------|
| Earnings (before interest) | \$400                      | \$1200                  | \$2000                   |
| Interest                   | $4000 \times 10\% = \$400$ | \$400                   | \$400                    |
| Net Income                 | 0                          | 800                     | 1600                     |
| EPS                        | 0                          | $\frac{800}{200} = \$4$ | $\frac{1600}{200} = \$8$ |

**Question:** What capital structure is “better”?

Suppose we have \$2000 and we want to invest in the levered company. We expect:

|          | Recession | Normal | Expansion |
|----------|-----------|--------|-----------|
| Earnings | 0         | 400    | 800       |

Suppose the firm remains unlevered. Can we generate the levered payoffs?

**Strategy:** Borrow \$2000 and use \$2000 of our own money to buy \$4000 of the unlevered firm's shares. Then, we have:

|                          | Recession | Normal | Expansion |
|--------------------------|-----------|--------|-----------|
| Earnings before interest | 200       | 600    | 1000      |
| Interest                 | 200       | 200    | 200       |
| Net Income               | 0         | 400    | 800       |

Suppose we have \$2000 and we want unlevered payoffs, but the firm becomes levered. We expect:

|          | Recession | Normal | Expansion |
|----------|-----------|--------|-----------|
| Earnings | 100       | 300    | 500       |

**Strategy:** Invest \$1000 in the levered firm and lend \$1000 at 10% interest. Then, we have:

|            | Recession | Normal | Expansion |
|------------|-----------|--------|-----------|
| Earnings   | 0         | 200    | 400       |
| Interest   | 100       | 100    | 100       |
| Net Income | 100       | 300    | 500       |

The firm is not adding any value by changing the capital structure.

**Definition 6.6 (Homemade Leverage).**

If a firm remains unlevered, but we want levered payoffs, we change our own  $D/E$  ratio to the desired level. This is called **homemade leverage**.

Recall that for the no tax case,

$$r_{WACC} = \frac{E}{V}r_E + \frac{D}{V}r_D.$$

By MM Proposition 1,

$$r_{WACC} = r_U = \text{unlevered rate of return} \implies r_U = \frac{E}{V}r_E + \frac{D}{V}r_D.$$

Then,

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

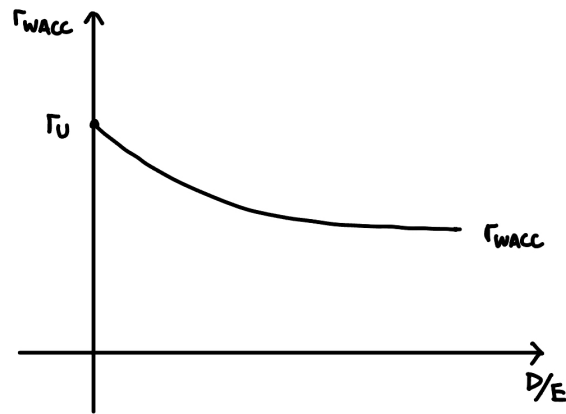
where  $r_U$  = cost of capital of the unlevered firm and  $r_E$  = cost of equity of the levered firm.

**Case 2:** With taxes.

**Proposition 6.2 (M & M Proposition 1 (With Taxes)).**

$V_L = V_U + DT_c$ , where  $D$  = debt, assumed to be in perpetuity.

*Note.* The value of the firm increases with leverage.



Higher  $D/E \implies$  lower  $r_{WACC} \implies$  higher firm value.

**Proposition 6.3 (M & M Proposition 2 (With Taxes)).**

$$r_E = r_U + \frac{D}{E}(r_U - r_D)(1 - T_c).$$

*Proof of MM 1 (with taxes).*

The firm makes \$EBIT/year in perpetuity. Then,  $D$  is also in perpetuity. So,

$$V_U = \frac{\text{EBIT}(1 - T_c)}{r_U}.$$



For the levered, bondholders receive  $r_D D$ . The shareholders receive  $(\text{EBIT} - r_D D)(1 - T_c)$ . So,

$$V_L = \text{PV of bondholders} + \text{PV of shareholders}.$$

The total payment is

$$(\text{EBIT} - r_D D)(1 - T_c) + r_D D = \text{EBIT}(1 - T_c) + r_D D T_c.$$

Thus,

$$V_L = \frac{\text{EBIT}(1 - T_c)}{r_U} + \frac{r_D D T_c}{r_D} = V_U + D T_c.$$

□

*Remark (Summary).*

- (1) Without taxes:  $D/E$  is irrelevant to firm value and  $r_{\text{WACC}}$  is constant.
- (2) With taxes: higher  $D \implies$  higher firm value and lower  $r_{\text{WACC}}$  as  $D/E$  increases.

---

### Lecture 17, 2025/11/10

---

Recall: We used homemade leverage to prove MM Proposition 1 (no taxes).

**Homemade Leverage:** create your own  $D/E$  ratio to match that of the firm.

The size of the cake is independent of who owns it.

Implications of MM Proposition 1 (with taxes):

- The value of the firm increases with debt.
- $r_{\text{WACC}}$  decreases as  $D/E$  increases (refer to the plot on the previous page).

#### Example (No Tax Case).

Suppose that ABC and XYZ are two identical firms, except that ABC is unlevered with \$750000 of stocks and XYZ is levered with  $D = \$375000$ ,  $E = \$375000$ . Also,  $r_D = 8\%$ , no taxes, and both firms have the same EBIT of \$86000/year in perpetuity.

**Q1:** Person A owns \$30000 of XYZ's stock. What return is this person expecting?

*Proof.* Note that EBIT = 86000. Then,

$$\text{Interest paid} = 0.08 \times 375000 = 30000.$$

So,

$$\text{Net earnings} = 86000 - 30000 = 56000.$$

Thus, A's share of the earnings is

$$\frac{56000}{375000} \times 30000 = \$4480.$$

A's rate of return is  $\frac{4480}{30000} = 14.93\%$ . □

**Q2:** Can A generate the same return using ABC's stock?

*Proof.* The strategy is to borrow \$30000 and use A's own \$30000 to buy \$60000 shares of ABC. Then, the earning from ABC is

$$\frac{60000}{750000} \times 86000 = \$6880.$$

Also,

$$\text{Interest on loan} = 0.08 \times 30000 = 2400.$$

So,

$$\text{Net Income} = 6880 - 2400 = 4480. \quad \square$$

**Q3:** What is  $r_U$  for ABC?

*Proof.* Note that

$$V_U = \frac{86000}{r_U} = 750000 \implies r_U = \frac{86000}{750000} = 11.47\%.$$

□

**Q4:** What is  $r_E$  for XYZ?

*Proof.* By MM Proposition 2 (no taxes),

$$r_E = r_U + \frac{D}{E}(r_U - r_D) = 11.47\% + 1 \times (11.47\% - 8\%) = 14.93\%. \quad \square$$

### 6.3 Limits to Debt

There are three main problems with too much debt:

- Risk of bankruptcy (already included in the stock price).
- Direct costs.
- Agency costs.

#### Direct Costs

- Lawyers' fees.
- Loss of reputation.

#### Agency Costs

The interest of the debtholders and that of the management may not be aligned (i.e. conflict of interest).

#### Example (Riskier projects may be accepted).

Suppose  $D = 100$ .

- Project 1: 50% chance of \$100 and 50% chance of \$200.
- Project 2: 50% chance of \$50 and 50% chance of \$240.

Note that project 1 should be chosen since it has a higher expected return. However, for project 1:

$$\text{Shareholders' payoff} = 0.5 \times 0 + 0.5 \times 100 = 50.$$

For project 2:

$$\text{Shareholders' payoff} = 0.5 \times 0 + 0.5 \times 140 = 70.$$

Even though project 1 is safer, we will choose project 2 since shareholders get a higher expected payoff.

**Example (Profitable Projects may be rejected).**

Suppose  $D = 4000$  and suppose the company has two possibilities:

- 50% chance of \$2400.
- 50% chance of \$100.

The available project has  $C_0 = -1000$  and  $C_1 = 1700$ . The project has to be financed through equity.

Without the project, the shareholders' expected payoff is

$$0.5 \times 0 + 0.5 \times 100 = 50.$$

With the project, the shareholders' expected payoff is

$$0.5 \times 100 + 0.5 \times 1800 = 950.$$

However, the initial cost of the project is \$1000 while the income increases by only \$950. So, the company will reject the project.

So having a lot of debt is not a good thing! And we don't know what the optimum  $D/E$  ratio is.

**Other Theories**

- **Free cash flow hypothesis:** more "free" cash  $\implies$  more waste  $\implies$  lower firm value.

So having debt  $\implies$  interest payments  $\implies$  less free cash  $\implies$  higher firm value.

- **Signaling theory:** taking debt  $\implies$  signal of that the company is doing well.

*Remark (Summary).*

- If the project looks like the assets of the company, use  $r_{WACC}$ .
- If the project is financed completely by equity, use  $r_E$  (from the SML).
- If the project is financed completely by debt, use  $r_D$ .
- If the project is risk-free, use  $r_f$ .

## A Tutorials

### A.1 Tutorial 1 (2025/09/19)

#### Mobius 2 Hints

Q1: Ignore CCA = cannot subtract depreciation.

Q10:  $\sigma(NPV)$  is the standard deviation of NPV.

Note. For  $Y = a + bX$ ,  $\sigma_Y = |b|\sigma_X$ .

#### Expected Utility

$U(X)$  = utility you get from lottery  $X$ . Under certain assumptions,

$$U(X) = \mathbb{E}[u(X)] = \sum u(x)f(x)$$

where  $u(x)$  is the von-Neumann Morgenstern utility function and  $f(x)$  is the PMF of  $X$ .

#### Risk Aversion

- A person is risk averse if  $u(\mathbb{E}[X]) > \mathbb{E}[u(X)]$  for all  $X$ .
- A person is risk neutral if  $u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$  for all  $X$ .
- A person is risk loving if  $u(\mathbb{E}[X]) < \mathbb{E}[u(X)]$  for all  $X$ .

Note.  $u(\cdot)$  is a concave function  $\iff$  risk averse.

**Example.** Let  $X$  be a lottery with PMF  $\mathbb{P}(X = 64) = \mathbb{P}(X = 8) = 0.5$  and let  $Y$  be a lottery with PMF  $\mathbb{P}(Y = 125) = \mathbb{P}(Y = 0) = 0.5$ . Consider  $u(x) = x^{1/3}$  and  $u(x) = \sqrt{x}$ .

Note that for  $X$  with  $u(x) = x^{1/3}$ ,

$$U(X) = 0.5 \cdot 64^{1/3} + 0.5 \cdot 8^{1/3} = 3.$$

For  $Y$  with  $u(x) = x^{1/3}$ ,

$$U(Y) = 0.5 \cdot 125^{1/3} = 2.5.$$

Find  $U(X)$  and  $U(Y)$  for  $u(x) = \sqrt{x}$ .

## Certainty Equivalent

Let  $CE(X)$  = amount of dollars for sure that makes you indifferent with the lottery  $X$ .

- Risk averse:  $CE(X) < \mathbb{E}[X]$ .
- Risk neutral:  $CE(X) = \mathbb{E}[X]$ .
- Risk loving:  $CE(X) > \mathbb{E}[X]$ .

### Definition A.1 (Certainty Equivalent).

The **certainty equivalent** of a lottery  $X$  is the amount of money  $CE(X)$  such that

$$u(CE(X)) = \mathbb{E}[u(X)].$$

If  $u(\cdot)$  is strictly increasing,  $CE(X) = u^{-1}(\mathbb{E}[u(X)])$ .

**Example.** Consider the following:

- Wealth: \$100.
- Fine: \$36 with probability  $p = \frac{1}{4}$ .
- M: \$9.25, parking fee to avoid the fine.
- Utility function:  $u(x) = \sqrt{x}$ .

*Proof.* Pay for parking =  $\sqrt{100 - M}$ . Don't pay for parking =  $\frac{1}{4}\sqrt{100 - 36} + \frac{3}{4}\sqrt{100} = 9.5$ .

So  $\sqrt{100 - M} > 9.5 \implies M < 100 - 9.5^2 = 9.75$ . So we should pay for parking.  $\square$

## Are We Rational?

### Example (Allais Paradox).

Consider the following choices:

- A: \$1 million dollars with  $p = 1$ .
- B: \$1 million dollars with  $p = 0.89$ , \$5 million dollars with  $p = 0.1$ , and \$0 with  $p = 0.01$ .
- C: \$0 with  $p = 0.9$ , \$1 million dollars with  $p = 0.1$ .
- D: \$0 with  $p = 0.89$ , \$1 million dollars with  $p = 0.11$ .

If you prefer A over B and C over D, you are not rational.

## **A.2 Tutorial 2 (2025/10/03)**